

Computer vision

Interpretability & Explainability

Responsible AI

Fairness

Contextual appropriate actions and equity

Explainability

Traceability and transparency in how the system works and why decisions are made

Trust

Confidence in the system to perform as intended and expected

Governance & Accountability

Ongoing monitoring and protections to meet these objectives



How to Enhance Trust and Clarity in Machine Learning

1

Prioritize explainability in model design



2

Ensure data quality and transparency



3

Adopt robust validation techniques



4

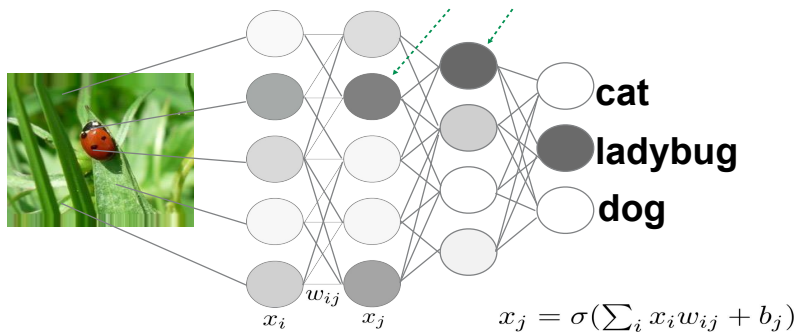
Design user-friendly interfaces



Explainability and Interpretability of AI

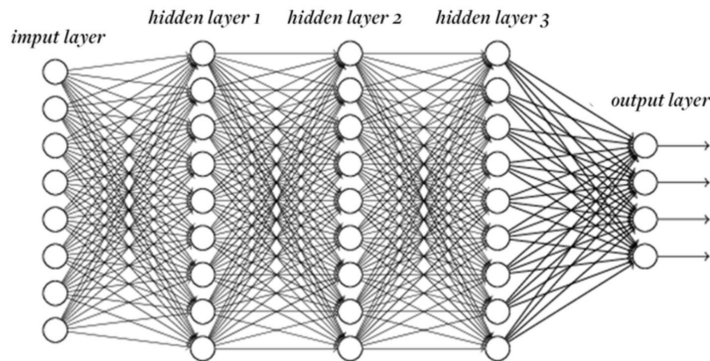
Explaining Neural Network Predictions

Classification → large activation



Because it performs hundreds of millions of calculations to get to that conclusion

-> poor transparency



Black box

Data
Data
Data

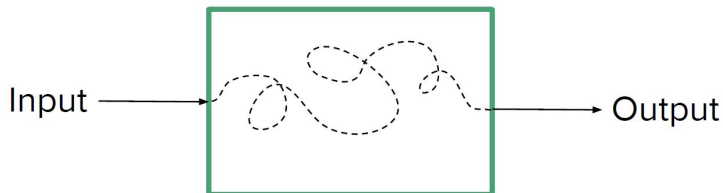


Black box:

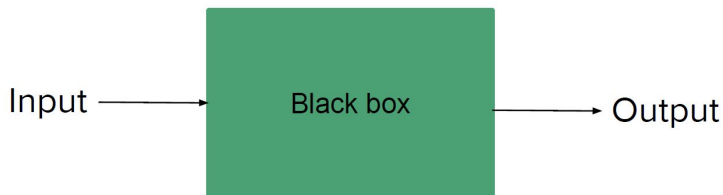
Statistical
procedures or
data science
algorithms



Results

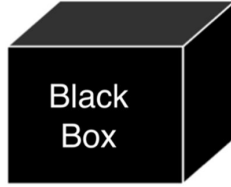
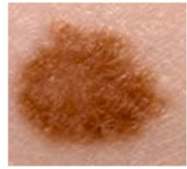


- Linear models
- Decision trees
-



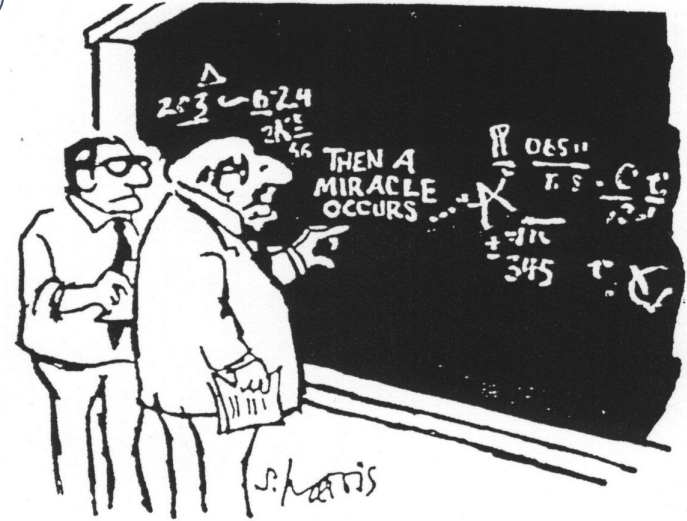
- DNN
- CNN
- RNN
-

Can we trust the black box?



diagnosis
??????????

Is the error minimization a
guarantee of correct
decision-making in practice?



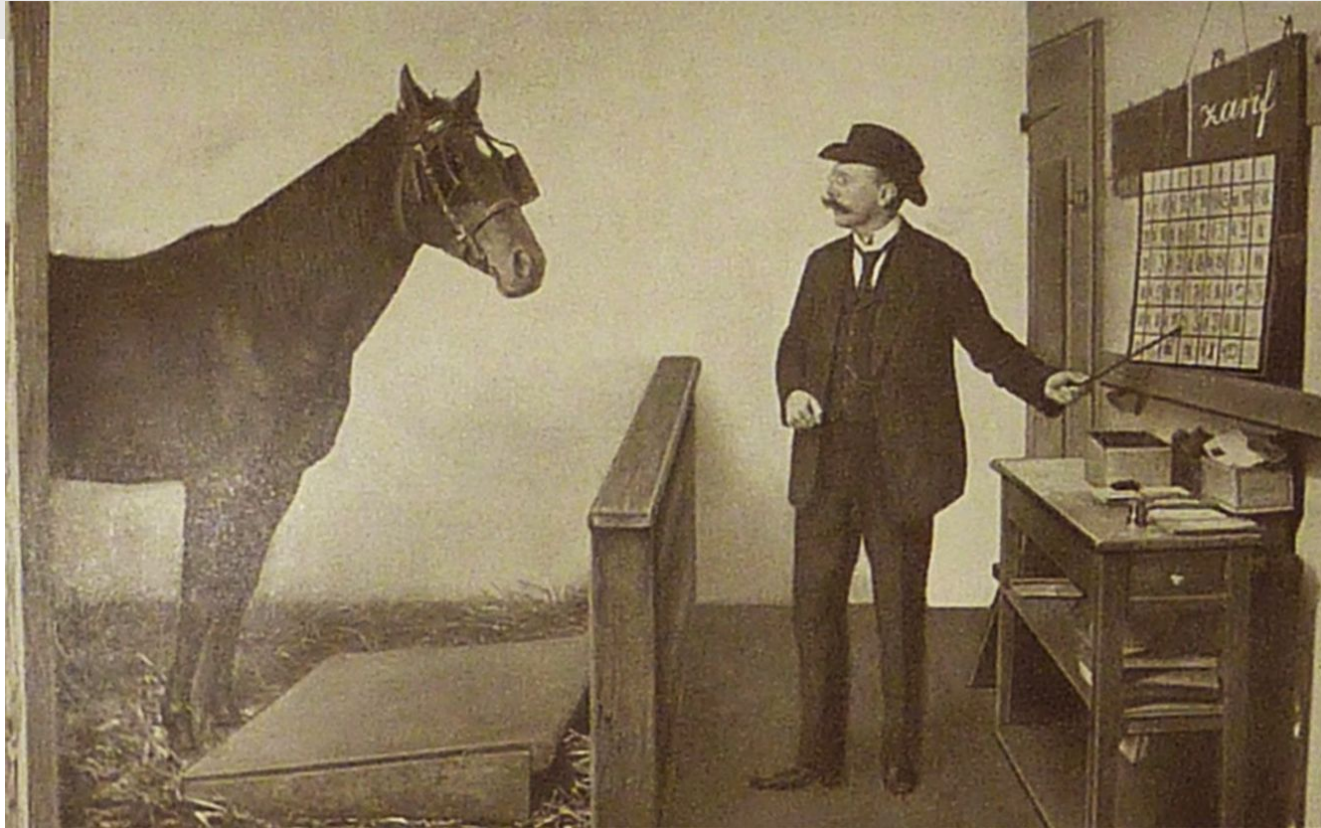
I think you should be a little
more specific, here in Step 2

„The Clever Hans effect“

was a horse that was claimed to have performed arithmetic and other intellectual tasks.

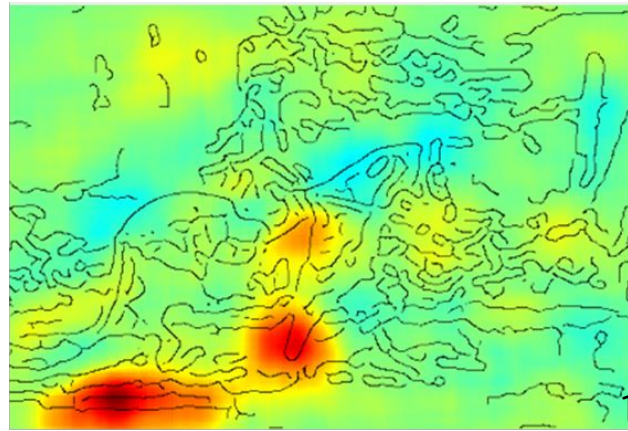
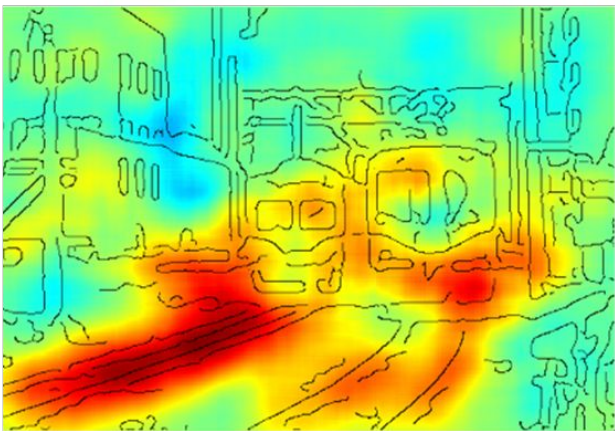
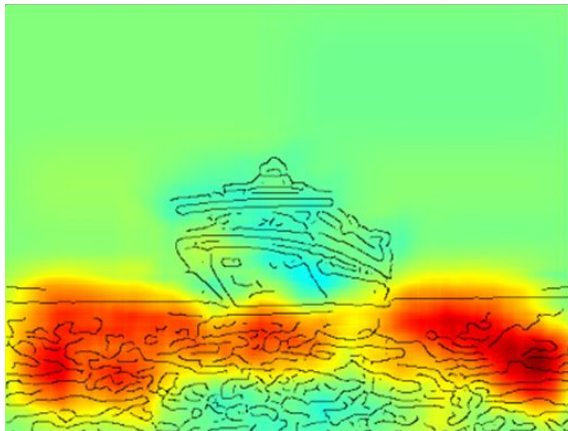


Unmasking Clever Hans Predictors



We have to guarantee that the network gives the right answers for the right reasons!

Unmasking Clever Hans Predictors



Object recognition example: Wolf vs. Husky



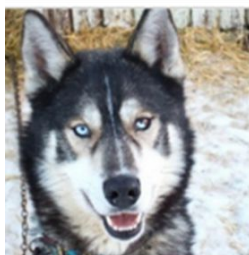
Predicted: **wolf**
True: **wolf**



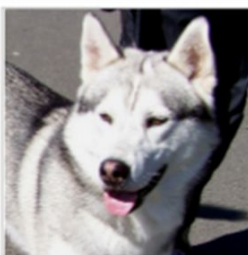
Predicted: **husky**
True: **husky**



Predicted: **wolf**
True: **wolf**



Predicted: **wolf**
True: **husky**



Predicted: **husky**
True: **husky**



Predicted: **wolf**
True: **wolf**

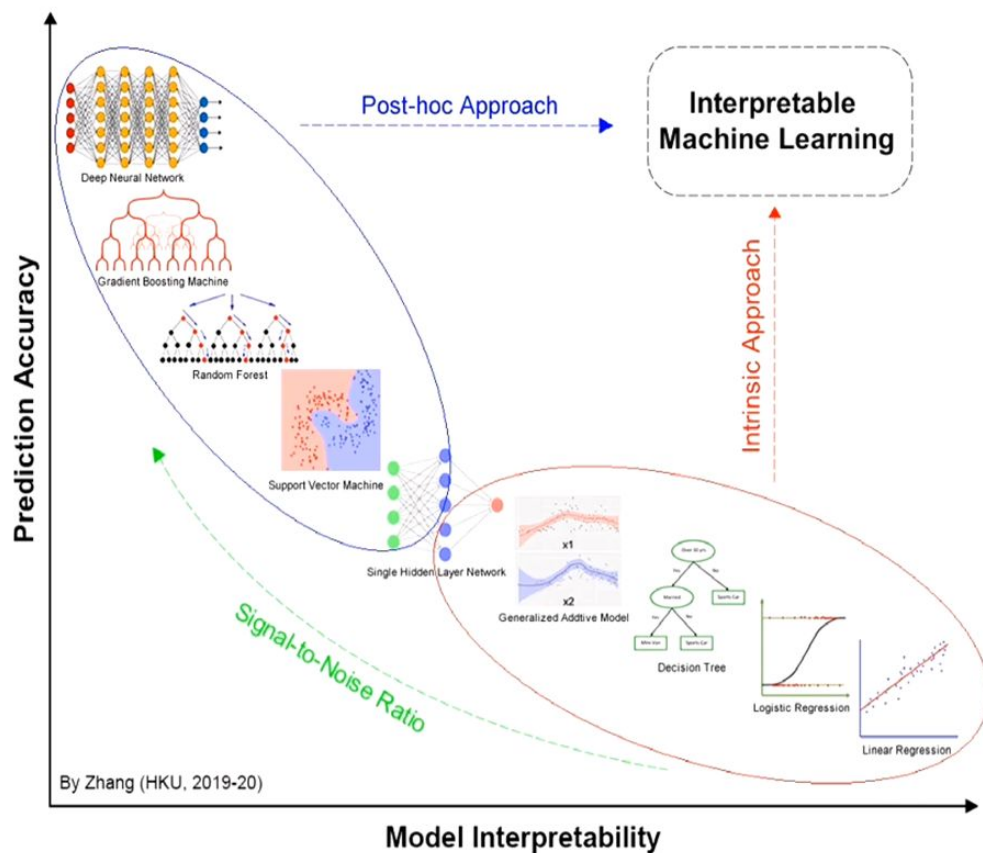
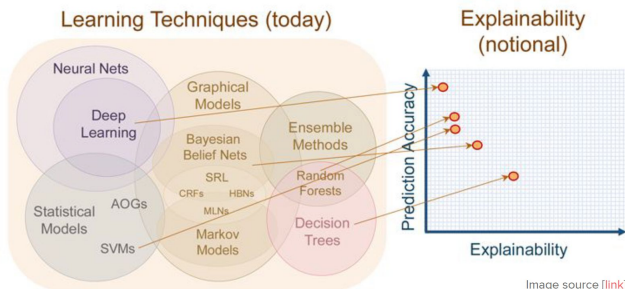
Interpretability , Explainability , Transparency

- **Interpretability:** It is defined as the ability to explain or to provide the meaning in understandable terms to a human.
- **Explainability:** Explainability is associated with the notion of explanation as an **interface between humans and a decision maker** that is, at the same time, both an accurate proxy of the decision maker and comprehensible to humans .
- **Transparency:** A model is considered to be transparent if by itself it is understandable.

Interpretability focuses on understanding the inner workings of the models, while explainability focuses on explaining the decisions made.

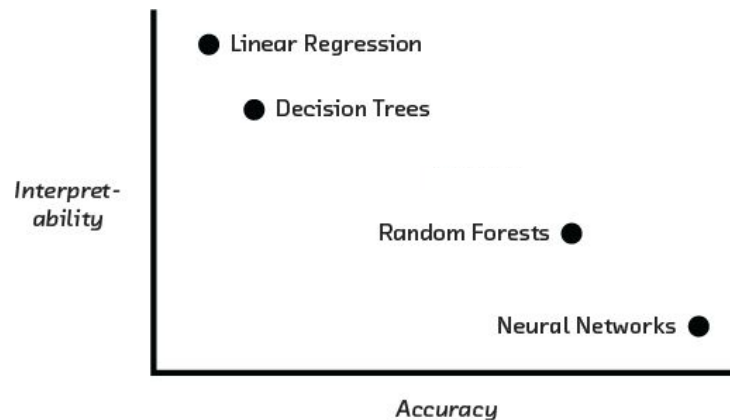
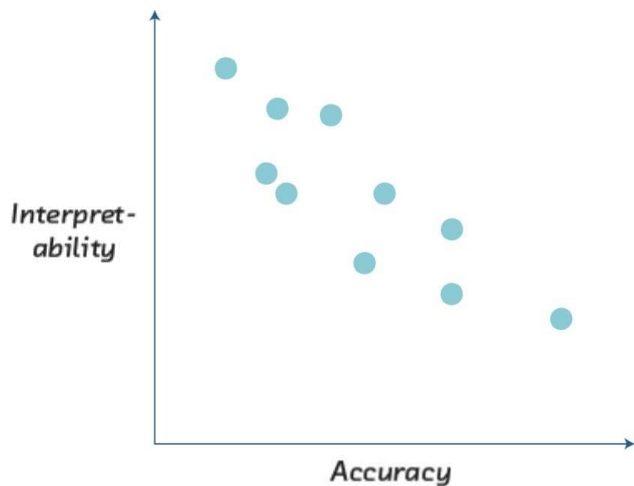
Interpretability & prediction accuracy

Accuracy vs interpretability



Inherently interpretable models vs Post-hoc explanations -> trade offs

Example



In *certain* settings, *accuracy-interpretability trade offs* may exist.

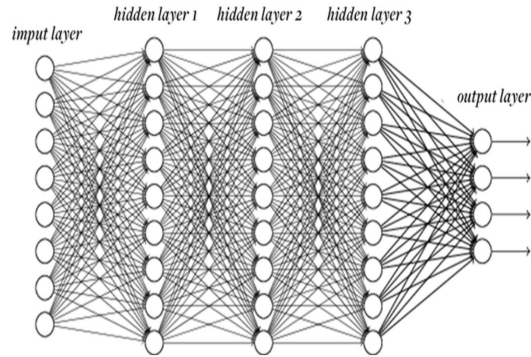
What is an Explanation?

Often the terms interpretability and explainability are used in the same sense and are not distinguished.

Simple def.:

Interpretable description of the model behavior

Classifier



trusted ?

Explanation

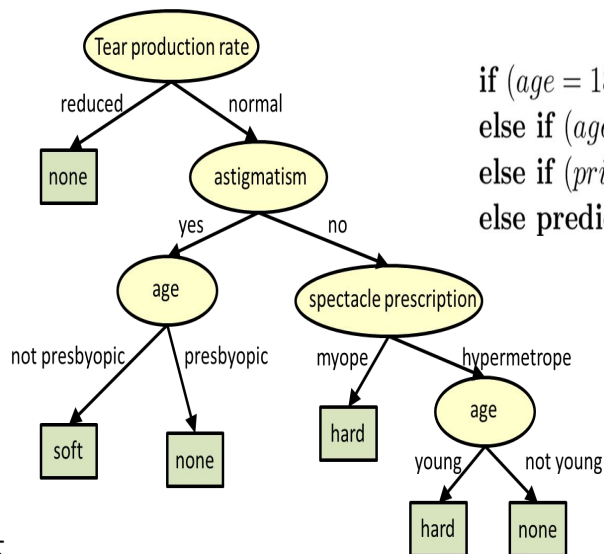
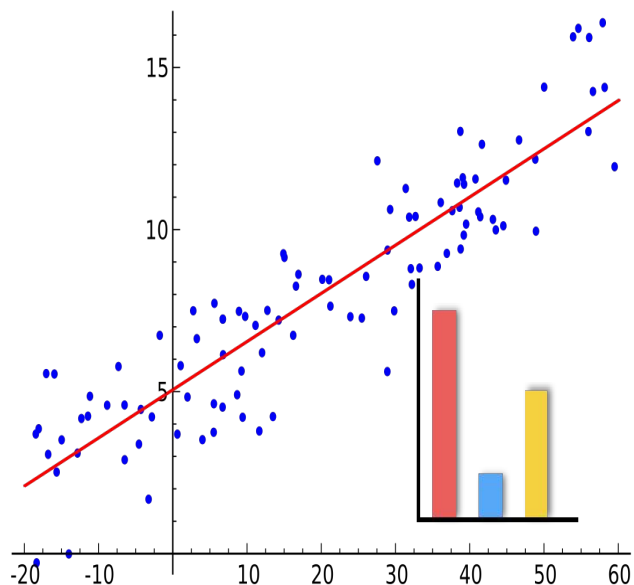
understandable

User



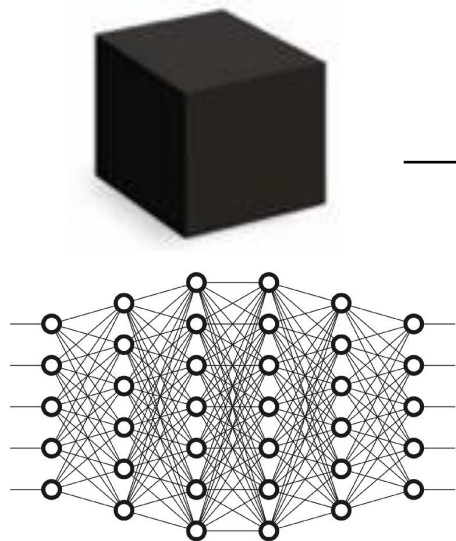
Inherently/directly
interpretable models vs.
post-hoc explanation

Build inherently interpretable predictive models

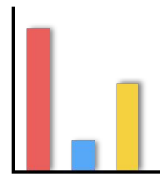


if ($age = 18 - 20$) and ($sex = male$) then predict *yes*
else if ($age = 21 - 23$) and ($priors = 2 - 3$) then predict *yes*
else if ($priors > 3$) then predict *yes*
else predict *no*

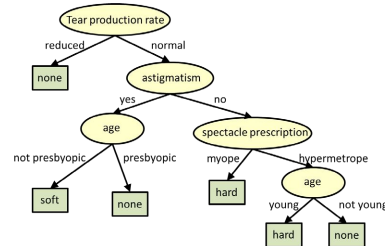
Explain pre-built models in a post-hoc manner



Explainer

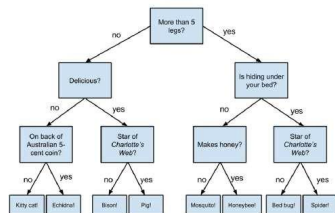


if (*age* = 18 – 20) and (*sex* = male) then predict *yes*
else if (*age* = 21 – 23) and (*priors* = 2 – 3) then predict *yes*
else if (*priors* > 3) then predict *yes*
else predict *no*



XAI

Inherently/ directly explainable model

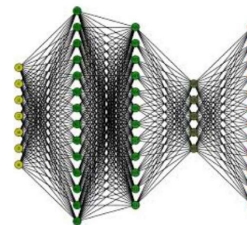


- Linear model
- Decision tree
- Rule-based model

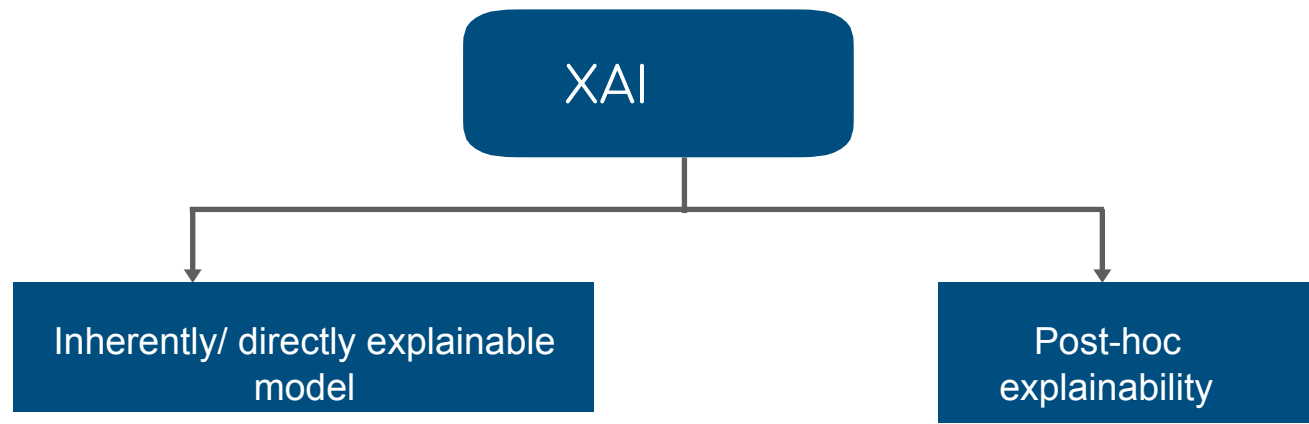
↓ Breaking the trade-off

- Generalized linear rule model
- Generalized additive models
- ...

Post-hoc explainability



- Deep neural networks
- Ensemble models



- Relatively simple frameworks
- The basis of their predictions can be intuitively understood
- Intrinsic interpretability is exhibited in varying degrees across different model families
- The most interpretable family of models include rule sets, decision trees and linear models.
- Owing to their simplicity they often yield modest or relatively low accuracies, particularly on complex tasks.

- Complex frameworks -> Neural Networks
- The key idea of post-hoc explainability is to explain model behavior **after the model has been learned.**
- However, the approach may compromise faithfulness as the explanation may not always follow the true rationale of predictions.

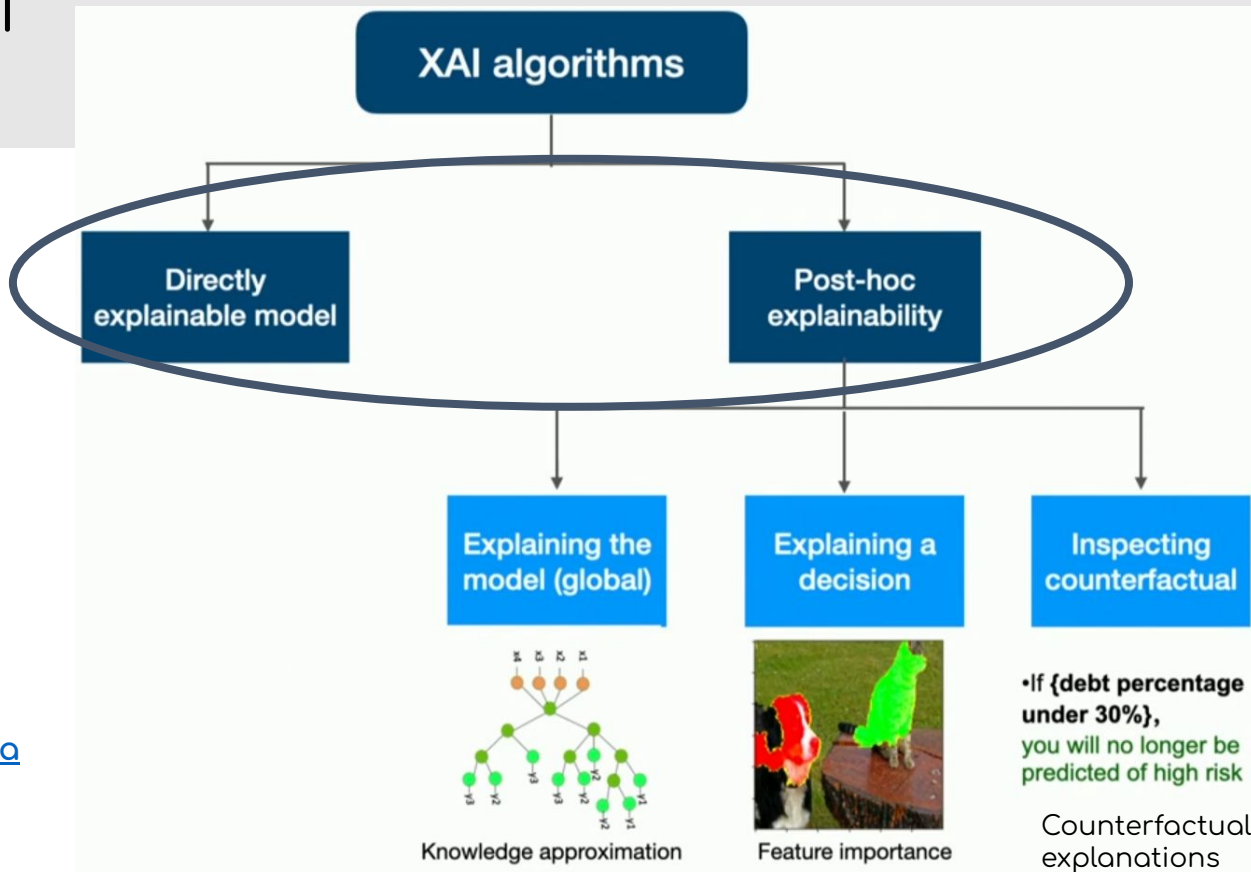
Explainable AI XAI

Overview of methods

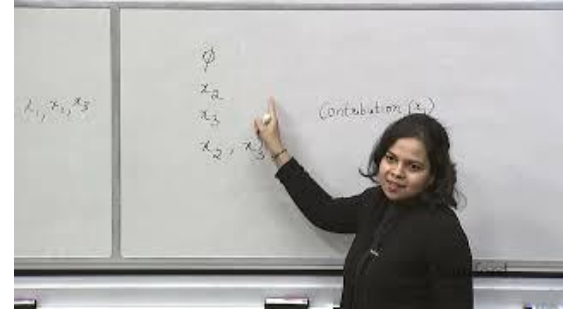
Stanford Seminar -
Human-Centered
Explainable AI: From
Algorithms to User
Experiences

Vera Liao

https://www.youtube.com/watch?v=ZxPV_KVq-tI&t=2890s



For more details, see our [CHI 2021 Tutorial on Intro to XAI](#)



Inherently/ directly explainable model

XAI methods

Intrinsically interpretable models

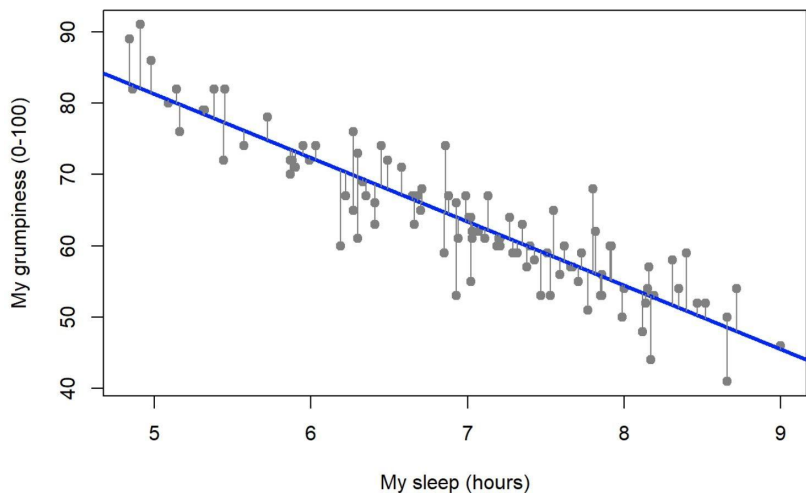
- Models that are interpretable by design
- No post-processing steps are needed to achieve interpretable.

Intrinsically interpretable model example: Linear Regression

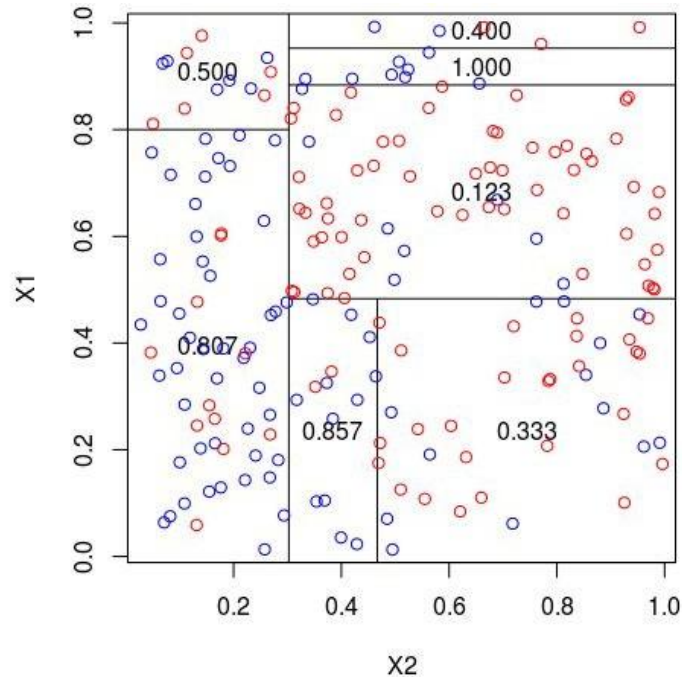
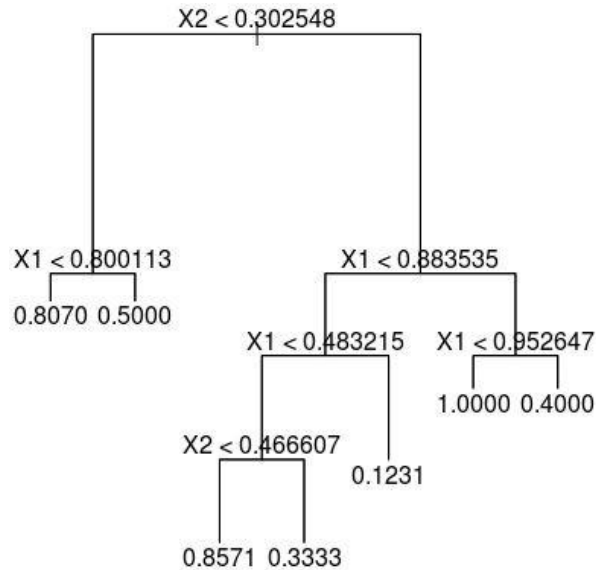
$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n,$$



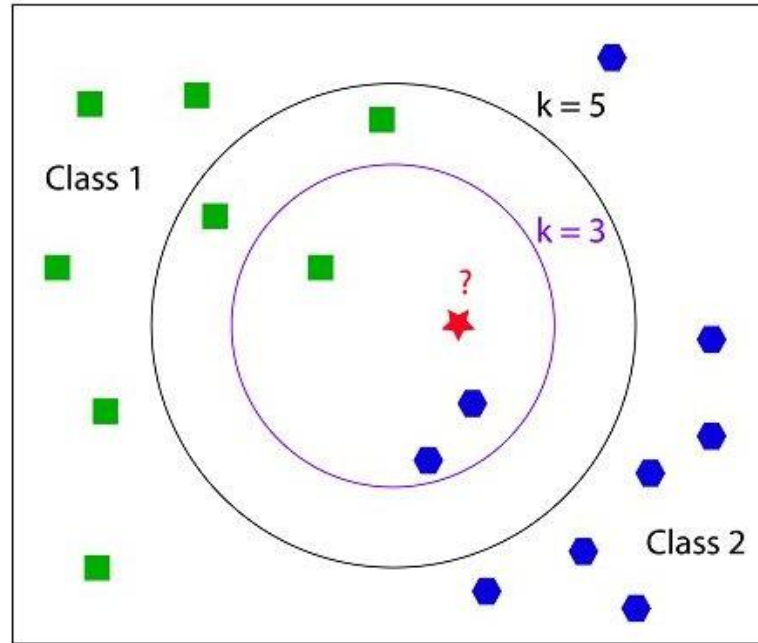
interpretable components



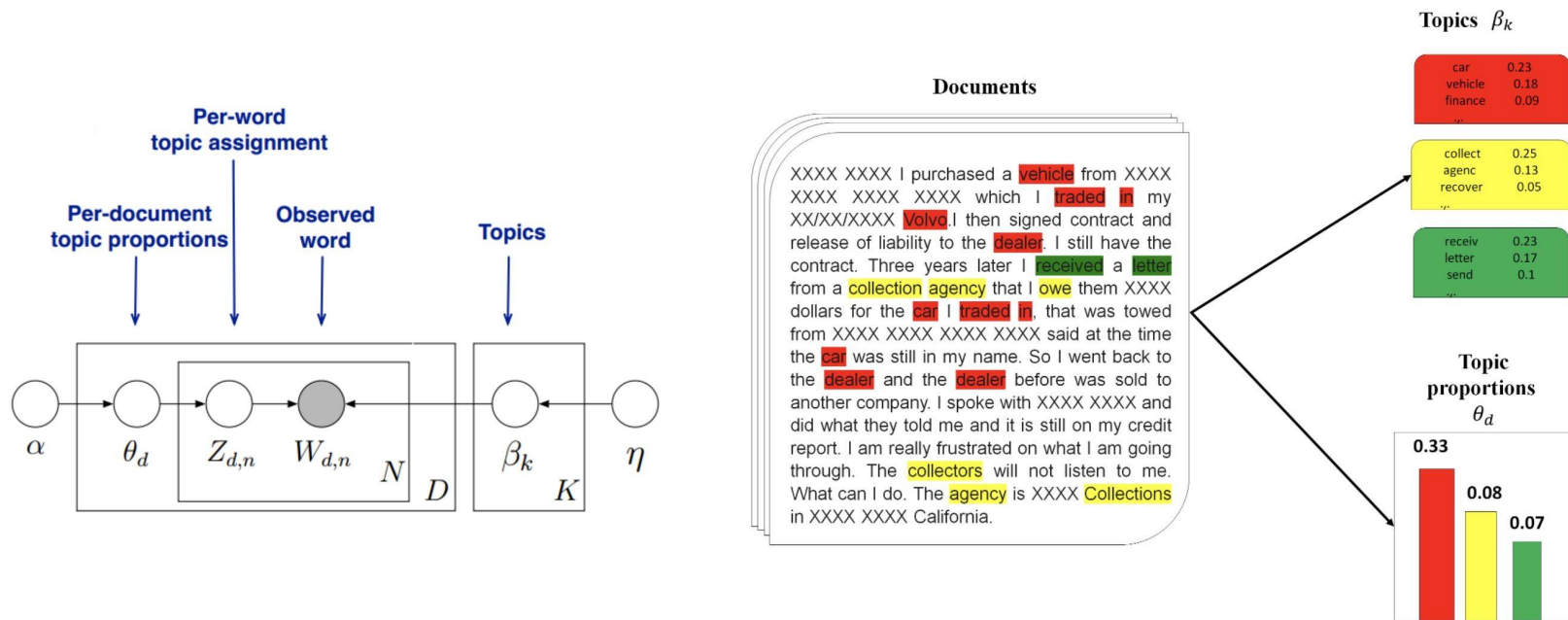
Intrinsically interpretable model example: Decision Tree



Intrinsically interpretable model example: K-Nearest Neighbors

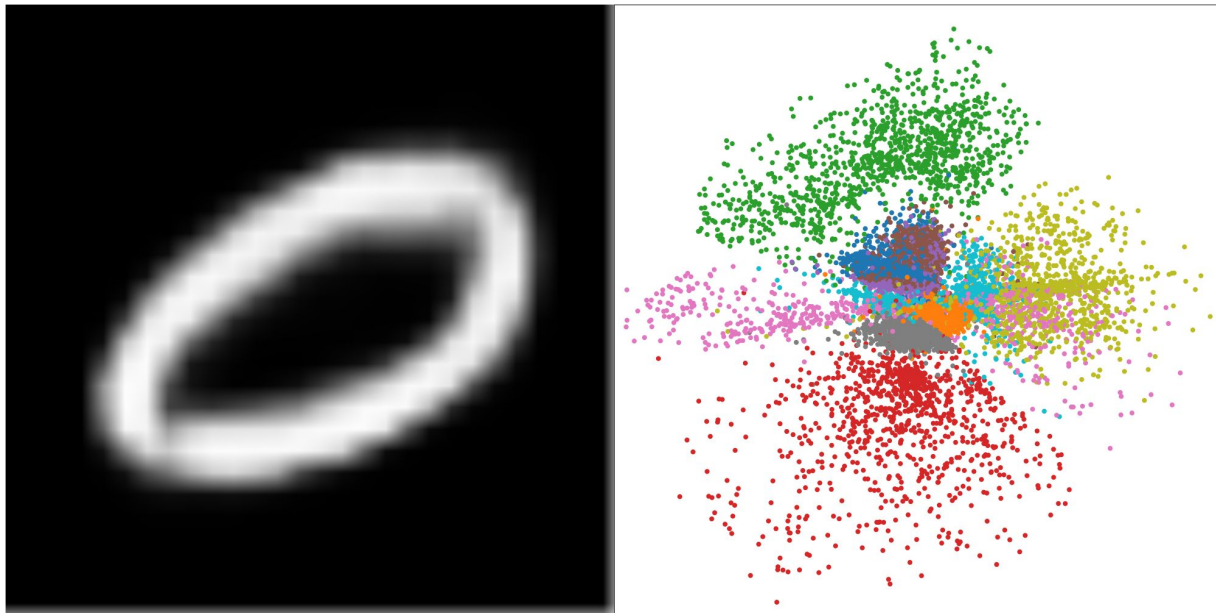


Intrinsically interpretable model example: Bayesian Models



Demo: VAE-Latent-Space-Explorer

<https://tayden.github.io/VAE-Latent-Space-Explorer/>



Post-hoc explainability

XAI methods

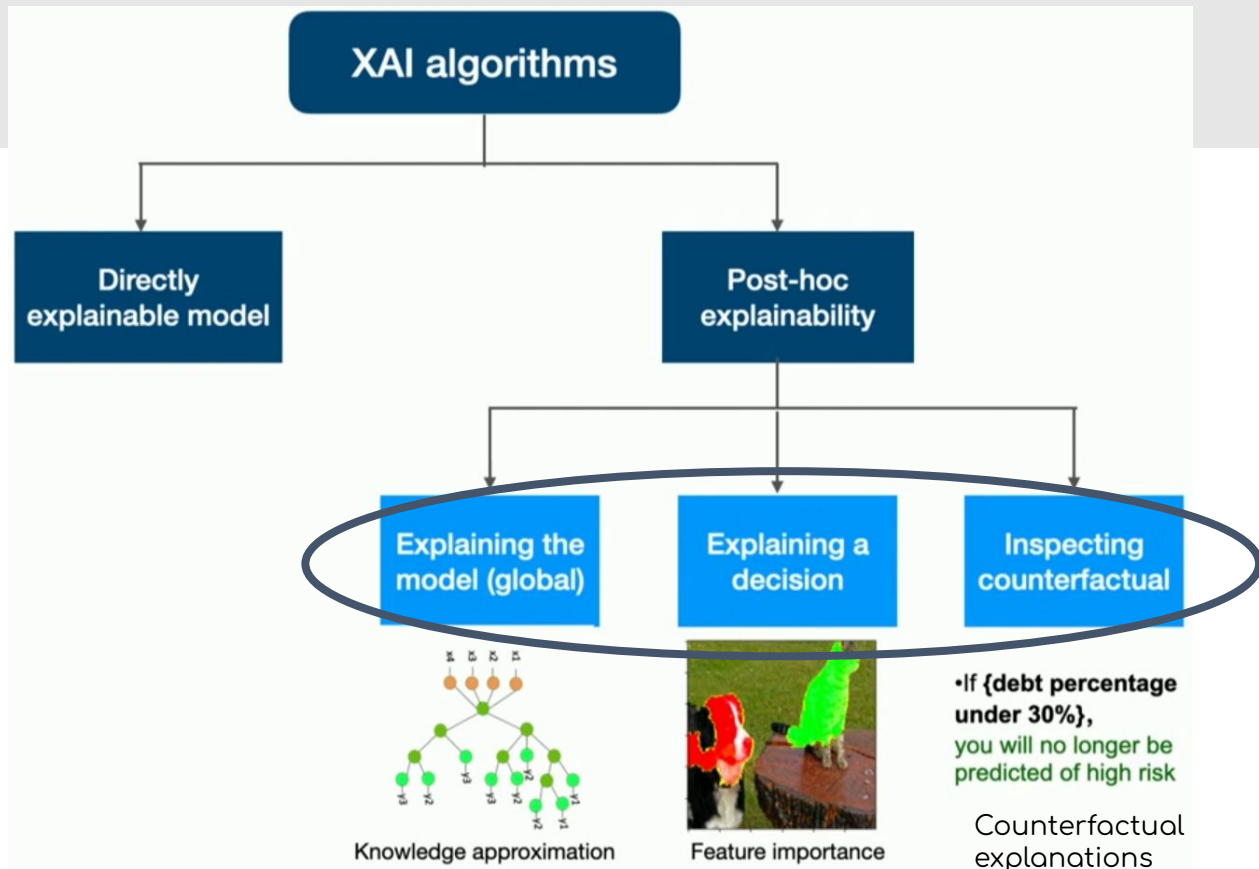
Explainable AI XAI

Overview of
methods

Stanford Seminar
- Human-Centered
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From Algorithms
to User
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https://www.youtube.com/watch?v=ZxPV_KVq-tI&t=2890s



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Global explanations vs. Local explanations

Local

Explaining Individual Predictions

Help unearth biases in the **local neighborhood** of a given instance
Help to verify if individual predictions are being made for the right reasons

vs

Global

Explain complete behavior of the AI model

Sheds light on **big picture biases** affecting larger subgroups
Help vet if the model, at a high level, is suitable for deployment

Methods of Global vs. Local Explanations

Methods of Local explanations :

- Feature relevance methods
- Visualizing Neural Networks (Local approach)
 - Saliency maps
- Prototypes/Examples

Methods of Global explanations :

- Collection of Local explanation
- Model distillation
- Representation based
- Visualizing Neural Networks (Global approach)

Agnostic vs. specific methods

Model-agnostic methods are versatile tools that work with any machine learning model by treating it as a black box.

Model-specific methods are designed for particular model architectures and use their internal structure for more precise but less flexible explanations.

Approaches of Local Explanations

Feature relevance methods

Lime & Shap

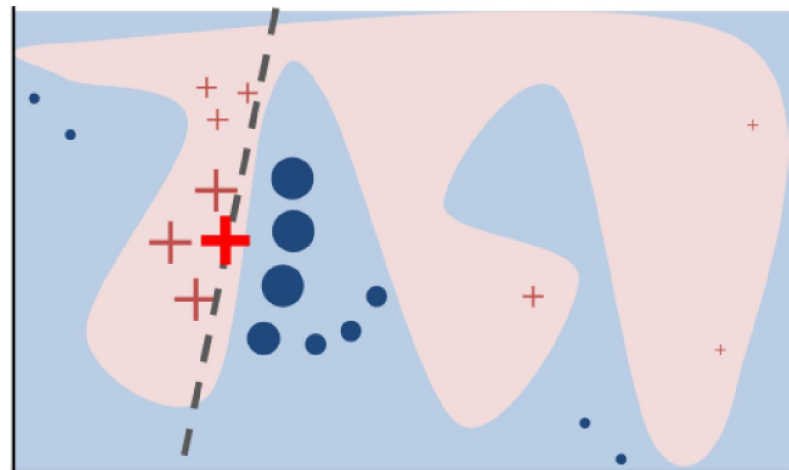
LIME – example

Toy example to present intuition for LIME from Ribeiro (2016).

The black-box model's complex decision function (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model.

The bold red cross is the instance being explained.

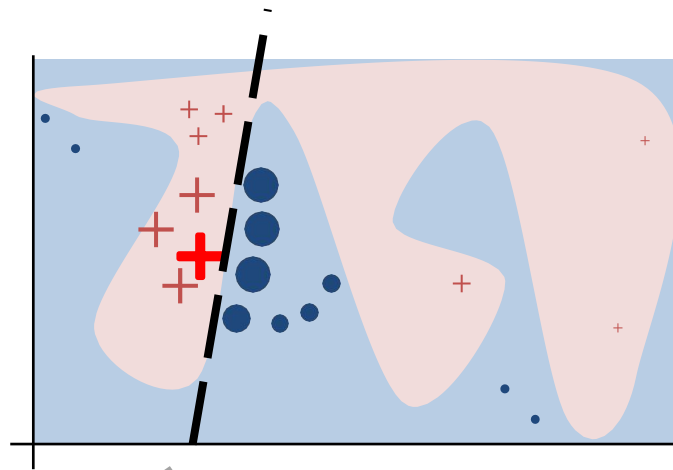
Local
Interpretable
Model-agnostic
Explanations



LIME: Local Interpretable Model-Agnostic Explanations

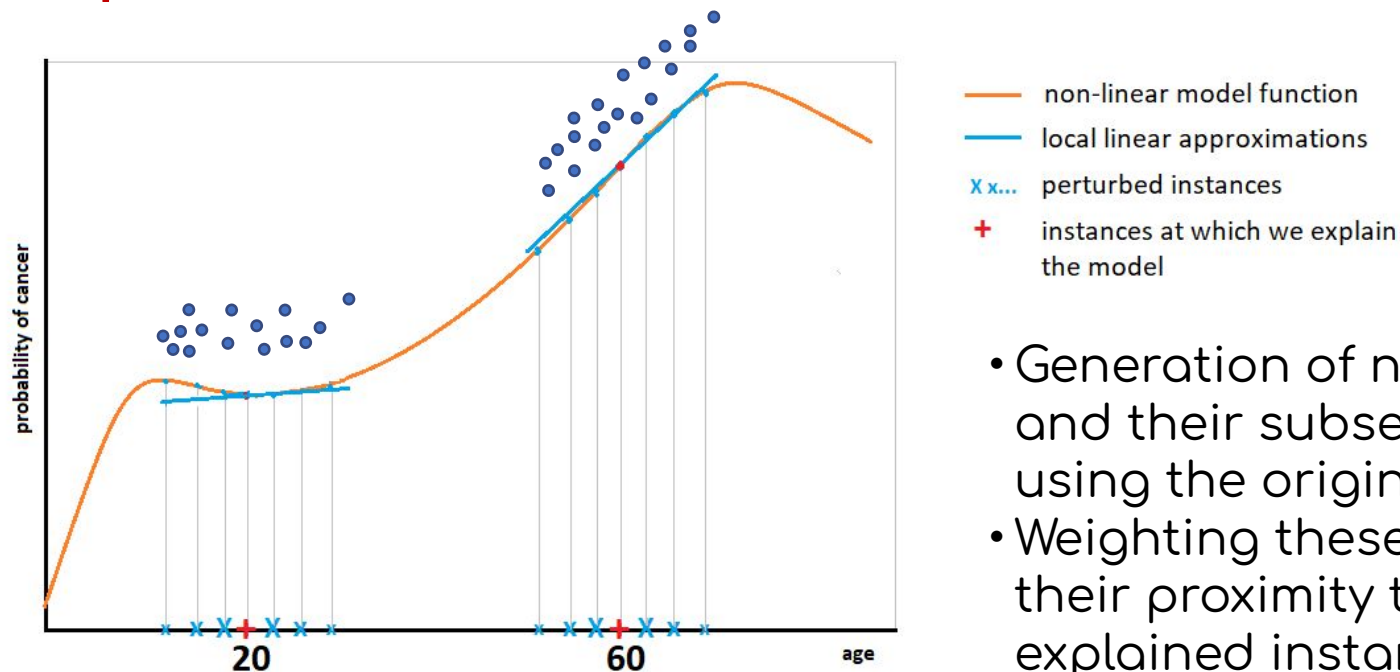
Steps:

1. Sample points around x_i
2. Use model to predict labels for each sample
3. Weigh samples according to distance to x_i
4. Learn simple linear model on weighted samples
5. Use simple linear model to explain



Another popular method which outputs feature importances: SHAP

Example of construction of LIME explanations for one attribute: age.



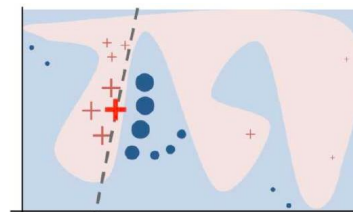
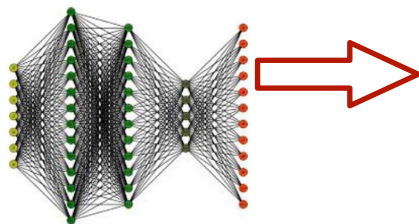
- Generation of new input samples and their subsequent prediction using the original model
- Weighting these new samples by their proximity to the given explained instances

<https://arxiv.org/pdf/1602.04938.pdf>

Ribeiro, M.T., Singh, S. and Guestrin, C., 2016, August. "Why should i trust you?" Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 1135-1144).

LIME is an agnostic method

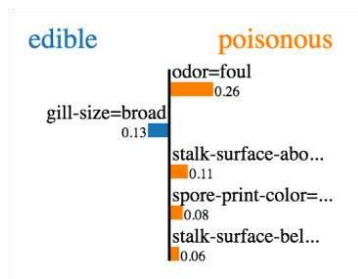
Various applications of LIME



LIME (Ribeiro et al. 2016)

Neural network, not directly explainable

Use a *post-hoc* XAI technique

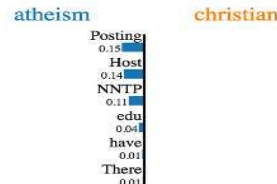


Tabular data

Images (explaining prediction of 'Cat' in pros and cons)



Image



Text with highlighted words

From: johnchad@triton.unm.edu (jchadwic)
Subject: Another request for Darwin Fish
Organization: University of New Mexico, Albuquerque
Lines: 11
NNTP-Posting-Host: triton.unm.edu

Hello Gang,

There have been some notes recently asking where to obtain the DARWIN fish. This is the same question I have and I have not seen an answer on the net. If anyone has a contact please post on the net or email me.

Texts

SHAP (SHapley Additive exPlanations)

SHAP (SHapley Additive exPlanations)

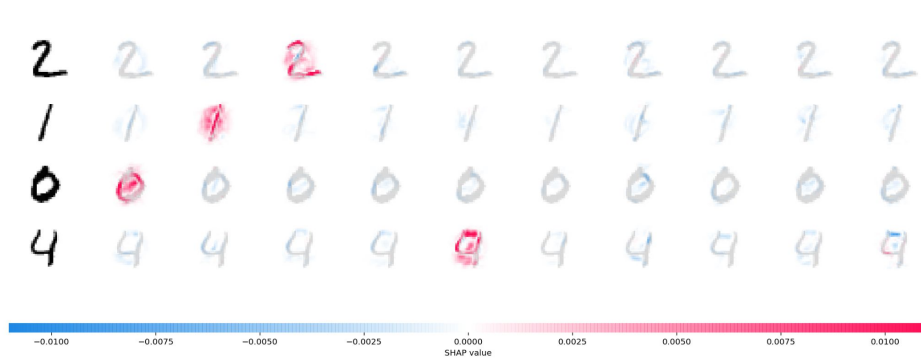
SHAP (SHapley Additive exPlanations) is a method to explain individual predictions.

SHAP is based on the game-theoretically optimal Shapley values.

SHAP has a strong mathematical background

Example of SHAP for image classification

The image explains ten outputs (digits 0-9) for four different images:



Red pixels increase the model's output while blue pixels decrease the output.

The input images are shown on the left, and as nearly transparent grayscale backings behind each of the explanations.

The sum of the SHAP values equals the difference between the expected model output (averaged over the background dataset) and the current model output.

Note that for the 'zero' image the blank middle is important, while for the 'four' image the lack of a connection on top makes it a four instead of a nine.

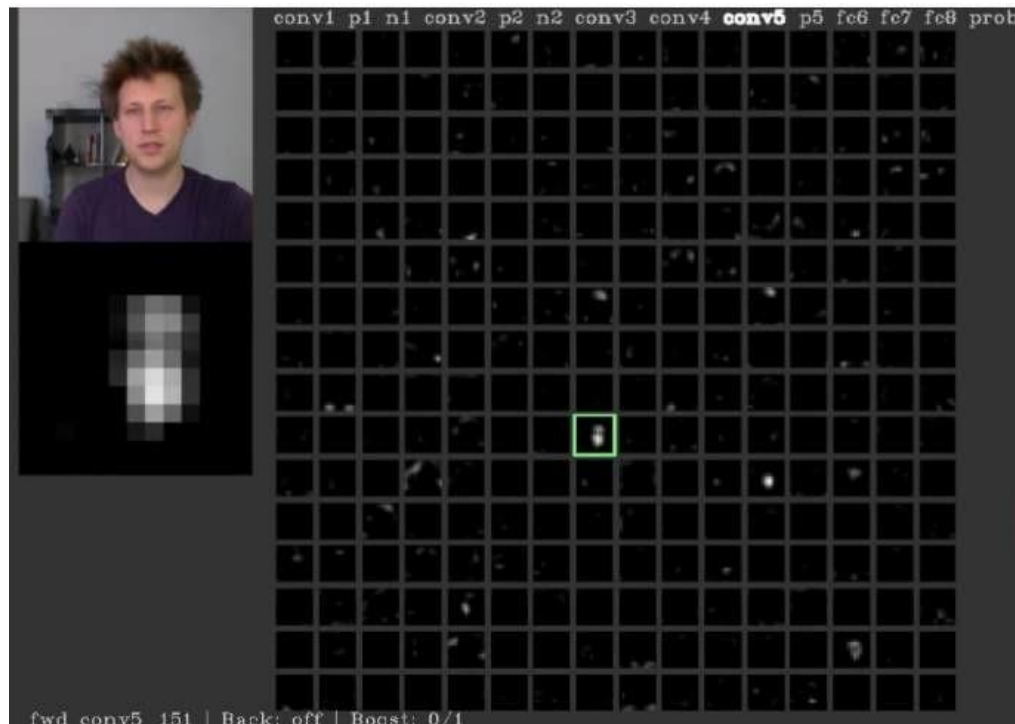
Visualizing Neural Networks (Local approach)

Saliency maps

Activation visualisation

Visualizing Activations

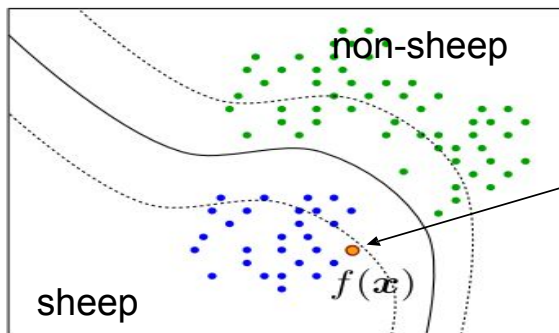
conv5 feature map is
128x13x13; visualize as 128
13x13 grayscale images



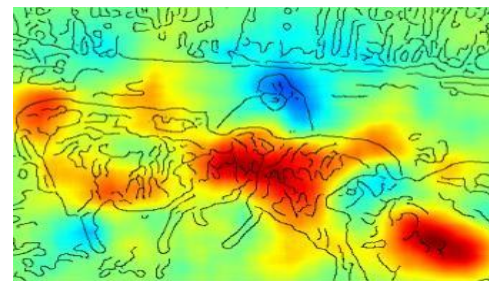
Yosinski et al, "Understanding Neural Networks Through Deep Visualization", ICML DL Workshop 2014.
Figure copyright Jason Yosinski, 2014. Reproduced with permission.

Saliency maps - Individual Explanations

“Why is a given image classified as a sheep?”



$$\text{heatmap} = LRP(x, f)$$



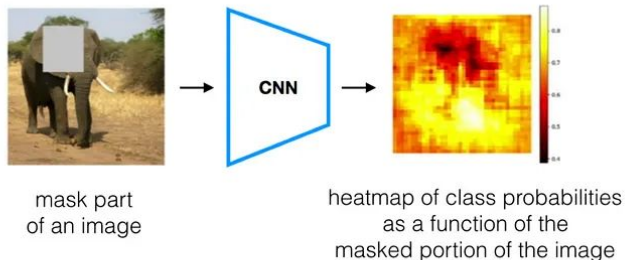
Images from **Lapuschkin'16**

Saliency via Occlusion Analysis/ Occlusion sensitivity

The occlusion analysis is a particular type of perturbation analysis where we repeatedly test the effect on the neural network output of occluding patches or individual features in the input image.

Problems/Challenges:

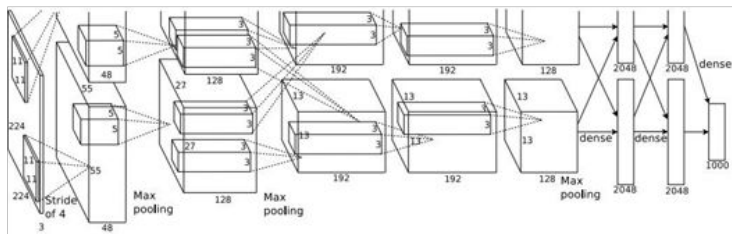
- time expensive method
- shape of the region?
- inpaint method?



Occlusion experiment with an image of an elephant.

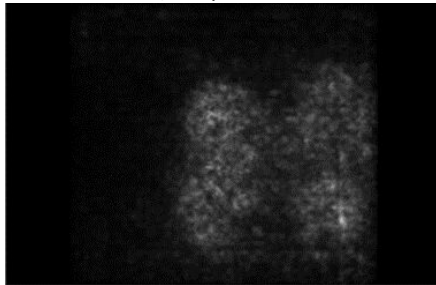
Saliency via Backprop

Forward pass: Compute probabilities



Dog

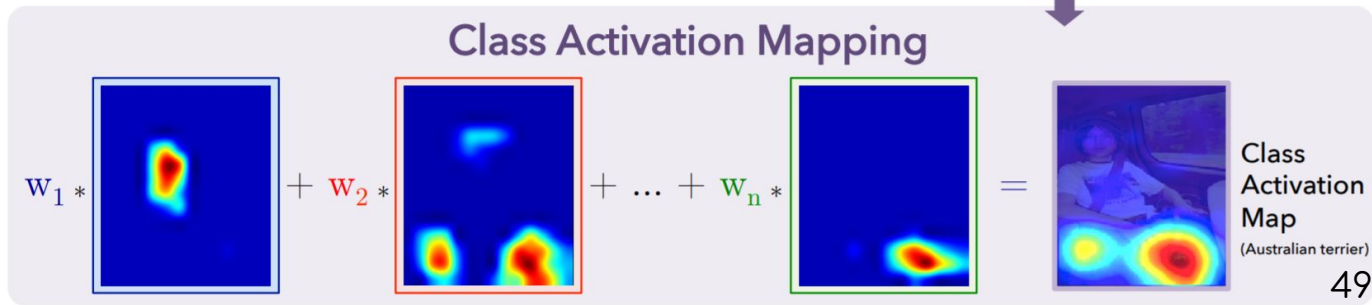
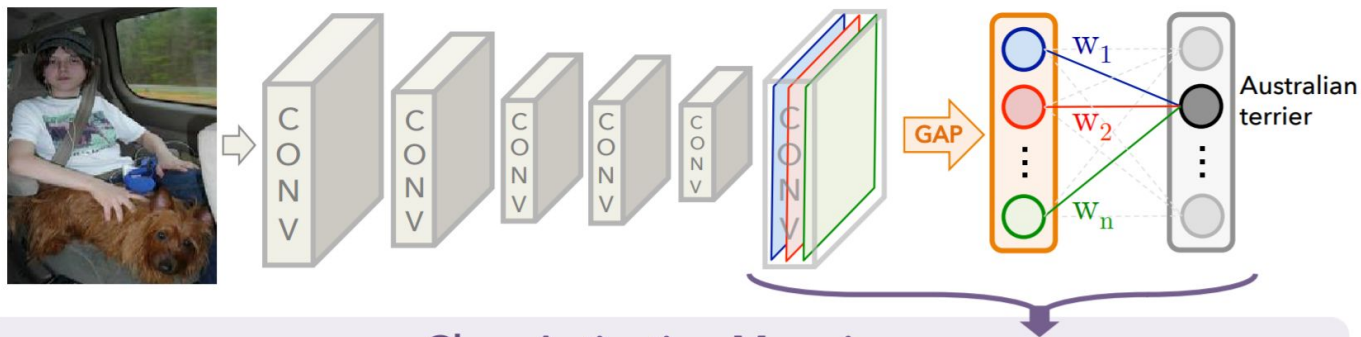
Compute gradient of (unnormalized) class score with respect to image pixels, take absolute value and max over RGB channels



Class Activation Maximization (CAM)

The Global Average Pooling (GAP) averages the activations of each feature map and concatenates and outputs them as a vector.

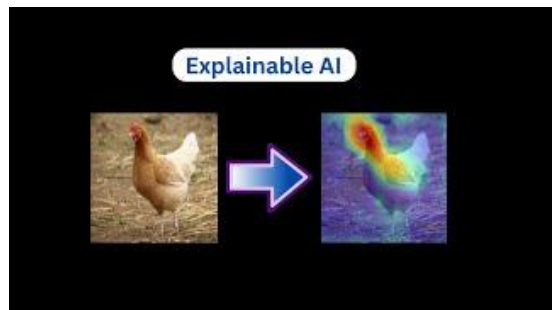
Then, a weighted sum of the resulted vector is fed to the final softmax loss layer.



CAM, GRAD-CAM, LRP

Gradient-weighted Class Activation Mapping (Grad-CAM)

- The main ideas of Grad-CAM are to visualize what parts of an image are important for a neural network prediction.
- Grad-CAM finds out which parts of an image most strongly affect a specific class prediction by combining the feature maps with how much they influenced the class score (via gradients).
- Provide class-specific heatmaps that show the regions most relevant to a specific output.



Gradient-weighted Class Activation Mapping (Grad-CAM)

To obtain the **class-discriminative localization map**, Grad-CAM computes the gradient of y_c (score for class c) with respect to feature maps A of a convolutional layer.

These gradients flowing back are global-average-pooled to obtain the importance weights α_k^c :

$$\alpha_k^c = \overbrace{\frac{1}{Z} \sum_i \sum_j}^{\text{global average pooling}} \underbrace{\frac{\partial y^c}{\partial A_{ij}^k}}_{\text{gradients via backprop}}$$

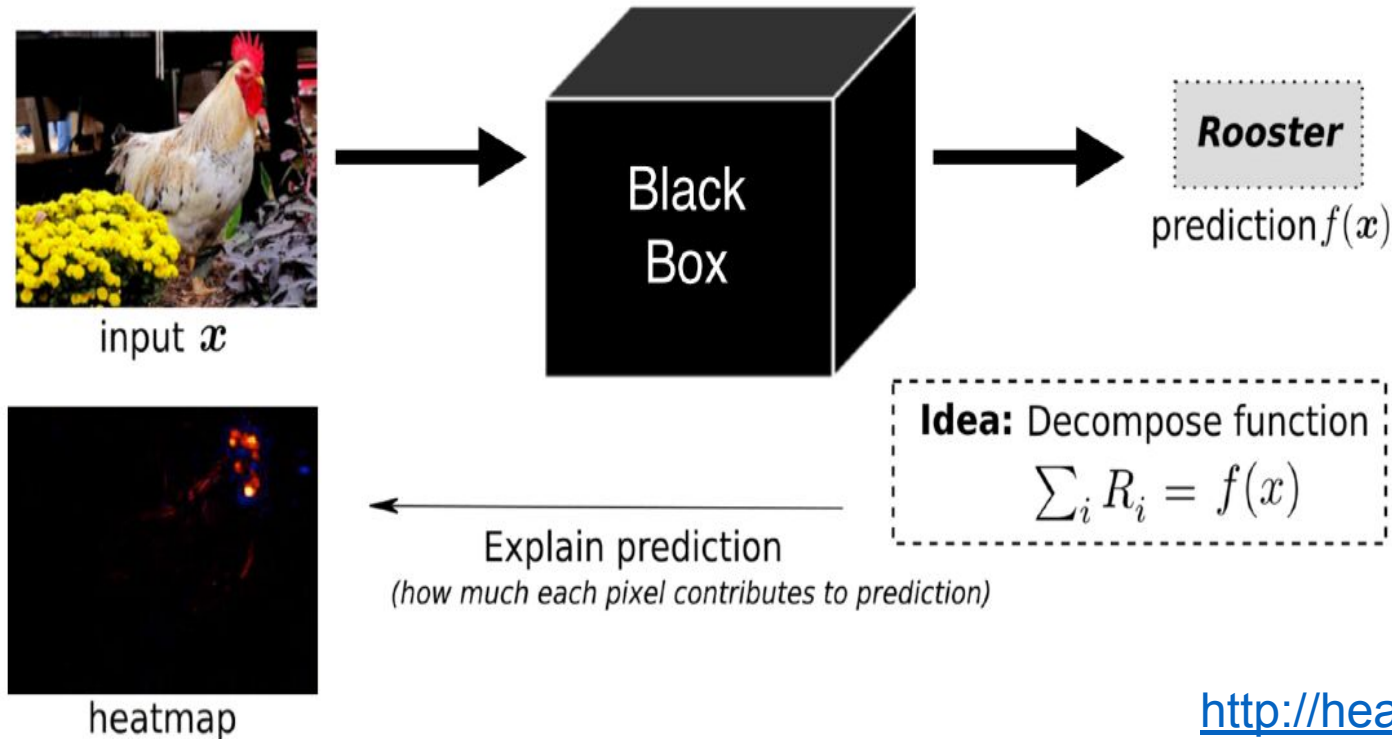
Grad-CAM heat-map is a weighted combination of feature maps, but followed by a ReLU:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\underbrace{\sum_k \alpha_k^c A^k}_{\text{linear combination}} \right)$$

Steps of Grad-Cam

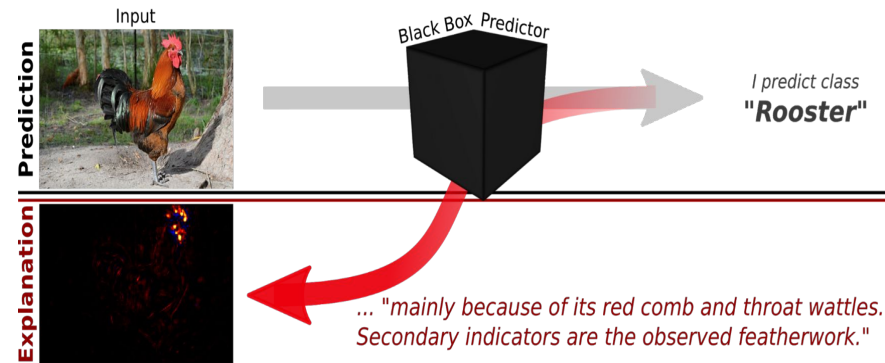
1. **Feed forward the image into the CNN** and choose the class you care about.
2. **Calculate** how important each feature map is for that class. Mathematically, this means taking **the gradient of the class output with respect to each feature map**.
3. **Compute importance weights**. For each feature map, you take the average of those gradients - this gives one number per feature map.
4. **Combine the maps**. You then make a weighted sum of all the feature maps - the more important maps contribute more to the final visualization.
5. **Keep only positive influence**. Apply a ReLU (Rectified Linear Unit) so that only areas that positively influence the “cat” prediction are kept (i.e., the pixels that support the decision).
6. **Resize and overlay**. The result is a coarse heatmap (red = important, blue = not important).

Saliency via Layer-wise Relevance Propagation (LRP)



<http://heatmapping.org/>

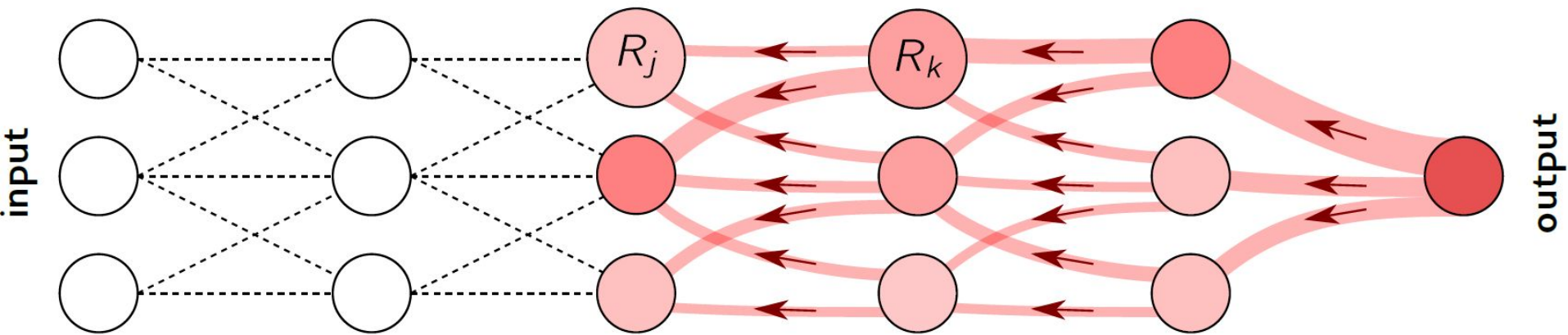
LRP Layerwise Relevance Propagation



[Demos](#)

- First, a standard forward pass through the model is performed, with which the network predicts.
- Then, the model's output is backward propagated layer-by-layer, following the neural pathways involved in the final prediction, by applying specific **LRP decomposition rules**.
- The result is a heatmap indicating the contribution of individual input features (e.g., of pixels) to the prediction, which can be computed for any hypothetically possible prediction.

Illustration of the LRP procedure

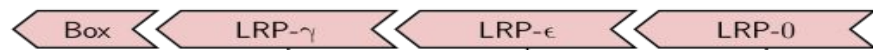
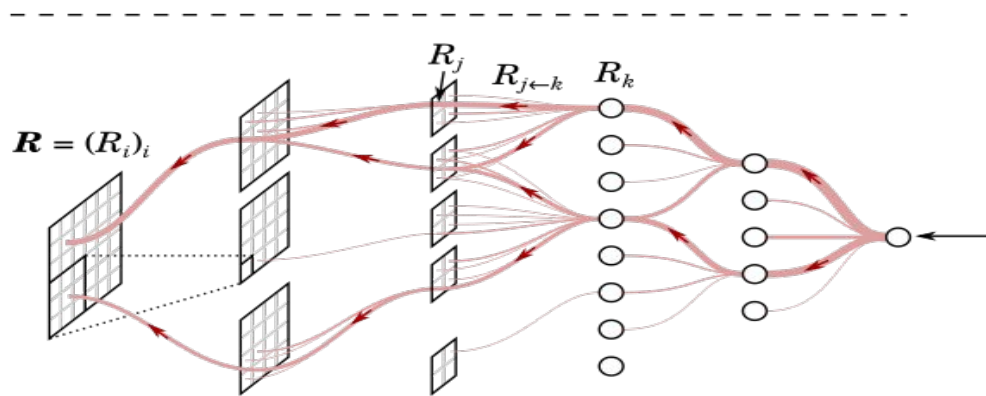
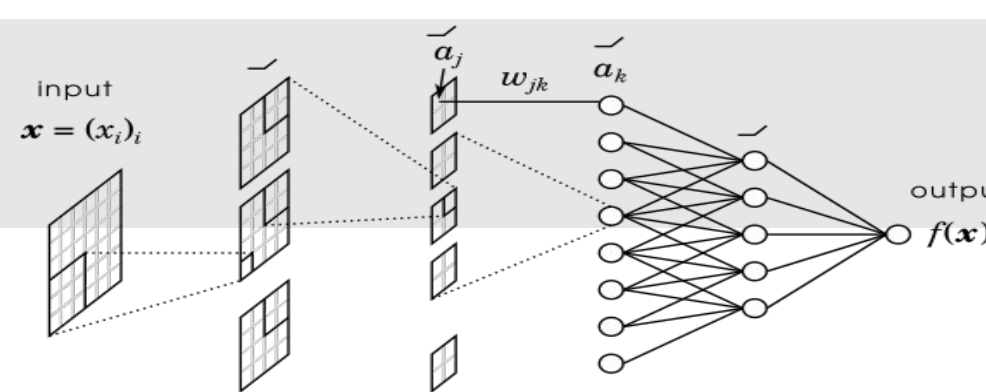


Each neuron redistributes to the lower layer as much as it has received from the higher layer.

Layer-wise Relevance Propagation (LRP)

Main idea:

- Predict relevance scores for input features using a classic forward pass.
- Backpropagate the network output according to LRP backpropagation rules
- LRP propagation rules can be improved for explanation quality – more sensitive at the top layers, more robust to noise at the bottom.



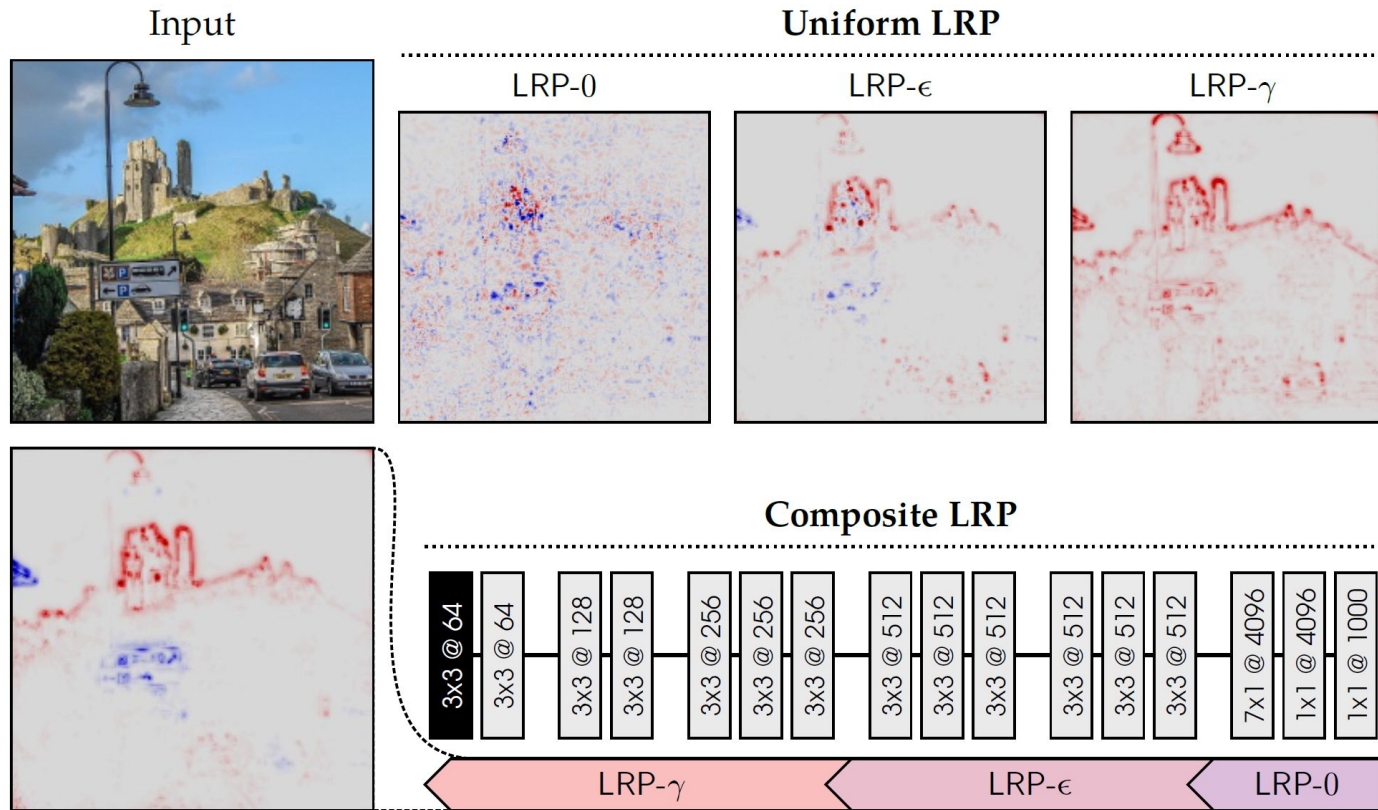
$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_{0,j} a_j w_{jk}} R_k$$

$$R_j = \sum_k \frac{a_j w_{jk}}{\epsilon + \sum_{0,j} a_j w_{jk}} R_k$$

$$R_j = \sum_k \frac{a_j (w_{jk} + \gamma w_{jk}^+)}{\sum_{0,j} a_j (w_{jk} + \gamma w_{jk}^+)} R_k$$

Various LRP rules

Input image and pixel-wise explanations of the output neuron 'castle' obtained with various LRP procedures.



Layer-Wise Relevance Propagation

Advantages

- Good quality explanations of deep neural networks.
- Fast : 1 forward and 1 reverse pass is enough.
- Flexible - with the help of several parameters, we can adjust the output so that it looks as good as possible.

Restrictions

- The LRP propagation strategy must be modified for each new architecture -> Other settings for Dense, CNN, ...
- LRP works based on some assumptions about the structure of the model - but that doesn't mean it works on all architectures.

<http://www.heatmapping.org/>

Age prediction: LRP heatmaps examples

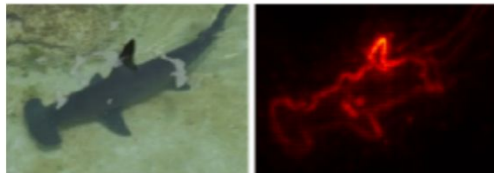
LRP heatmaps demonstrating the effects of ImageNet pre-training (middle) compared to additional IMDB-WIKI pretraining (bottom).

All heatmaps show the model decision with respect to age group (60+).

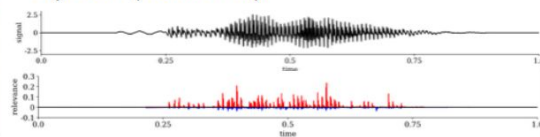


LRP examples

General Images (Bach' 15, Lapuschkin'16)



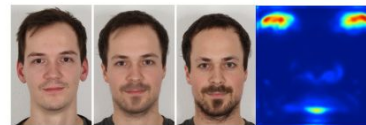
Speech (Becker'18)



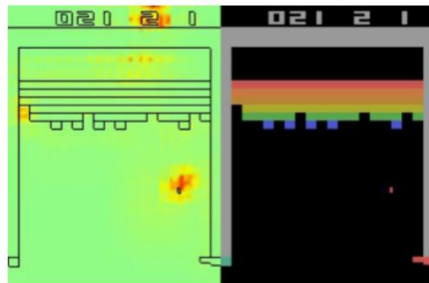
Text Analysis (Arras'16 &17)

do n't waste your money
neither funny nor susper

Morphing Attacks (Seibold'18)

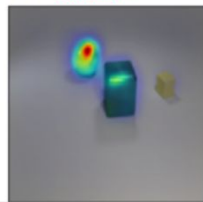


Games (Lapuschkin'19)

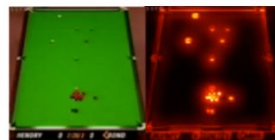


VQA (Samek'19)

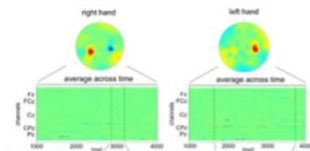
there is a metallic cube ; are
there any large cyan metallic
objects behind it ?



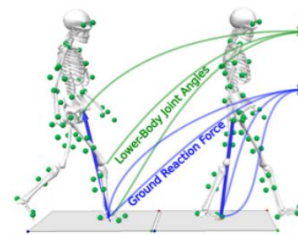
Video (Anders'19)



EEG (Sturm'16)



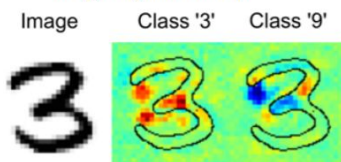
Gait Patterns (Horst'19)



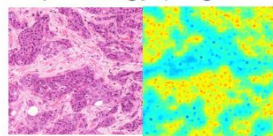
Faces (Lapuschkin'17)



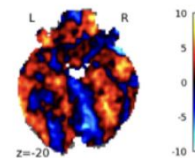
Digits (Bach' 15)



Histopathology (Hägele'19)

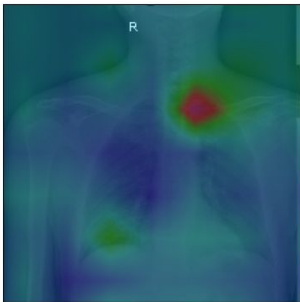
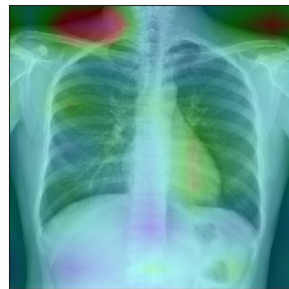
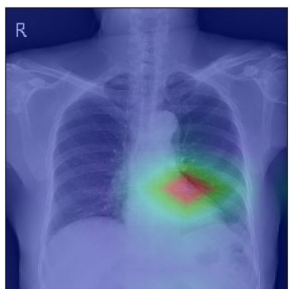


fMRI (Thomas'18)



Unmask “Clever Hans” examples

Lung disease classification

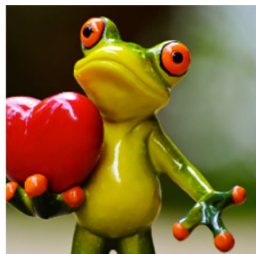
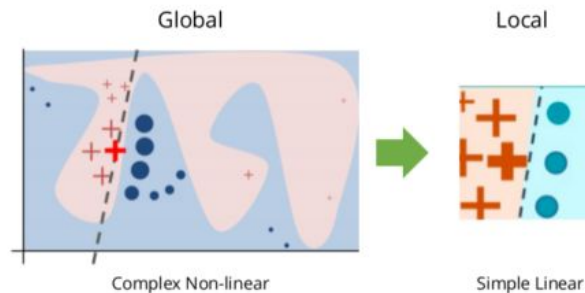


Lime visualisation in an image

LIME

used for the image classification

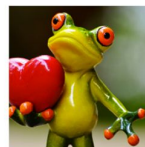
- local
- agnostic



Original Image



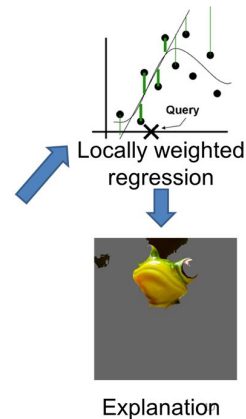
Interpretable Components



Original Image
 $P(\text{tree frog}) = 0.54$



Perturbed Instances	$P(\text{tree frog})$
	0.85
	0.00001
	0.52



Prototypes/Examples

Prototypes/Example explanation

Use examples (synthetic or natural) to explain individual predictions

Influence Functions (Koh & Liang 2017)

Identify instances in the training set that are responsible for the prediction of a given test instance

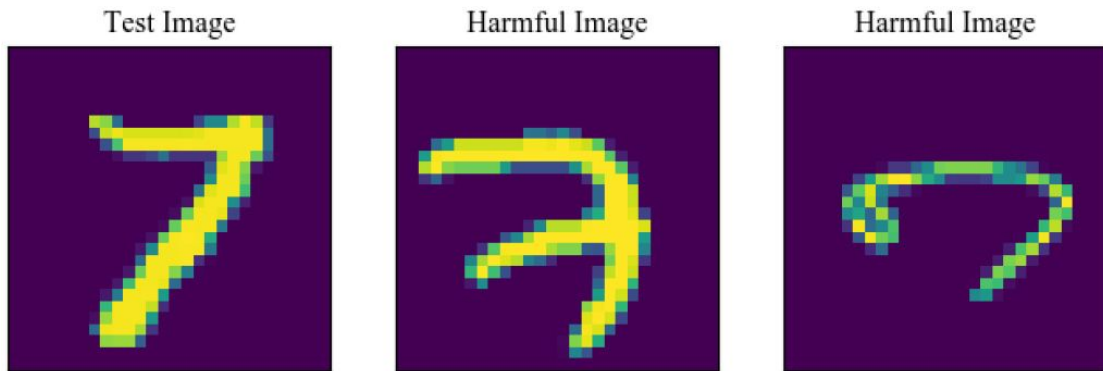
Activation Maximization (Erhan et al. 2009)

Identify examples (synthetic or natural) that strongly activate a function (neuron) of interest

Influence Functions

Given a prediction for one sample, do other samples in the dataset have positive effects or negative effects on that prediction?

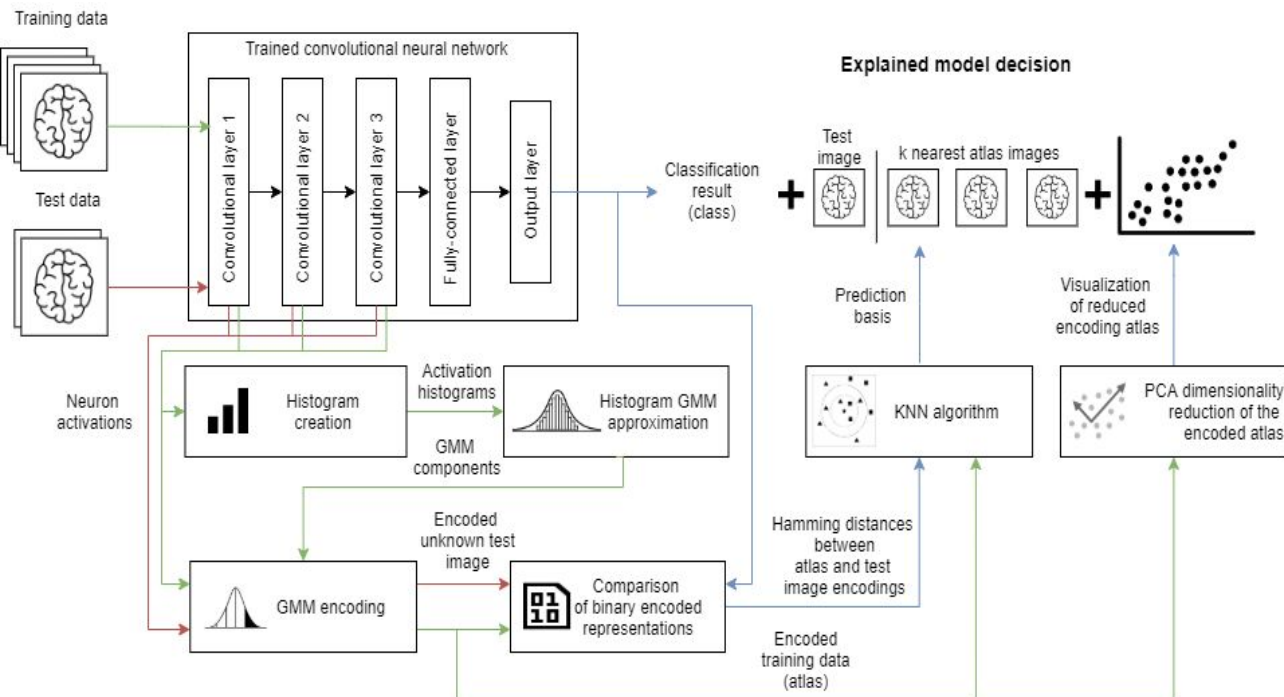
This analysis could also help identify mis-annotated labels and outliers existing in the data. As Fig. shows, given a LeNet-5 like network, two harmful images for a given image are identified by the influence function.



Based on the influence function, two harmful images that have the same label as the test image are identified.

Similar examples

Example

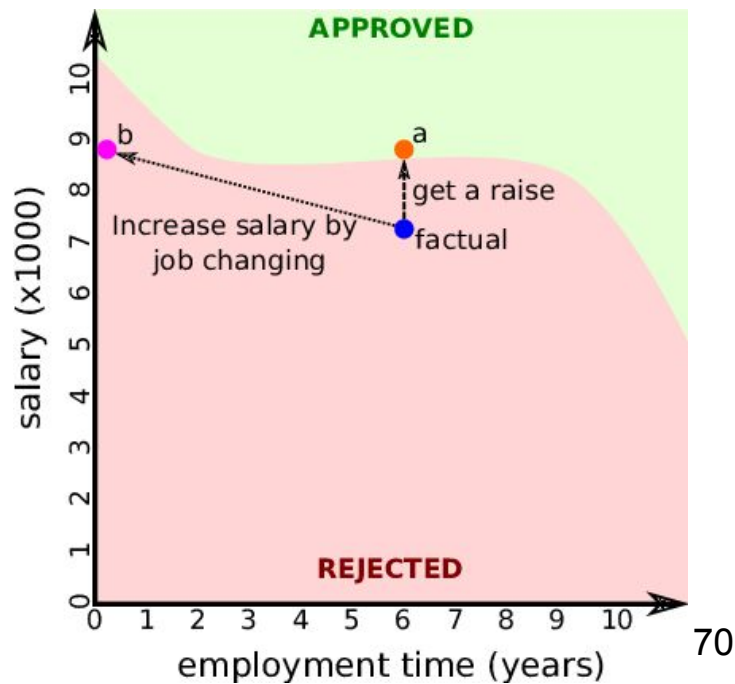


encoding of neuron
signal activities using
Gaussian Mixture
Model (GMM)
method of atlas
creation: a set of
representative
samples to support
explanations of future
predictions
-> matching process

Counterfactuals

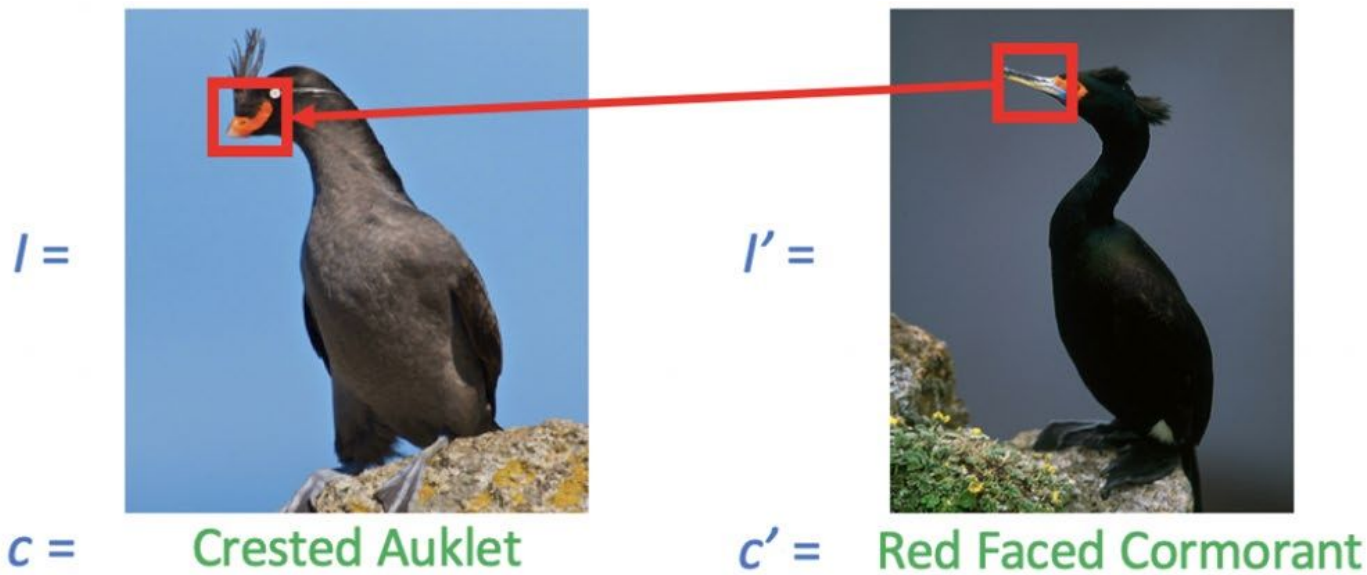
Counterfactual explanation

A counterfactual explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output.



Counterfactual explanation

What features need to be changed and by how much to flip a model's prediction?



Example

Why is this a cardinal, but not a scarlet tanager?



Predict Candidate Counterfactual Evidence

Counter-Class: Scarlet Tanager



Explanation
Generator

This is a *Scarlet Tanager*
because it is a *red bird* with a
pointy beak and *black eyes*.
...
This is a *Scarlet Tanager*
because it is a *red bird* with
black wings and a *pointy beak*.

Evidence Checker



Evidence Checker

Attribute	Score
red bird	0.94
pointy beak	0.87
black eyes	0.92
black wings	0.12

Counterfactual Explanation Generator



It is not a *Scarlet Tanager* because it does not
have *black wings*.

Adversarial Examples

Adversarial examples



“panda”

57.7% confidence

$+ .007 \times$



noise

$=$

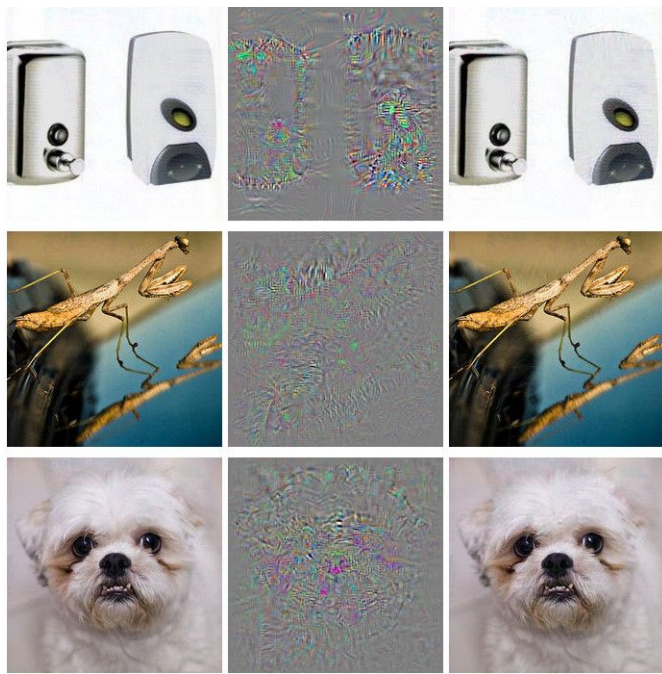


“gibbon”

99.3% confidence

Adversarial examples

Adversarial examples are counterfactual examples with the aim to deceive the model, not interpret it.



Adversarial examples for AlexNet.
All images in the left column are correctly classified.

The middle column shows the (magnified) error added to the images to produce the images in the right column all categorized (incorrectly) as “Ostrich”.

Loss function to generate adversarial examples

These adversarial examples were generated by minimizing the following function with respect to :

$$\text{loss}(\hat{f}(\mathbf{x} + \mathbf{r}), l) + c \cdot |\mathbf{r}| \quad \text{where:}$$

$\mathbf{x}$ is an image (represented as a vector of pixels),

$\mathbf{r}$ is the change to the pixels to create an adversarial image ($\mathbf{x} + \mathbf{r}$ produces a new image),

l is the desired outcome class,

and the parameter c is used to balance the distance between images and the distance between predictions.

- the first term is the distance between the predicted outcome of the adversarial example and the desired class l
- the second term measures the distance between the adversarial example and the original image.

One pixel attack

A one-pixel attack is a type of adversarial attack that modifies a single pixel in an image to fool a deep neural network, causing it to misclassify the image with high confidence.

Examples of one pixel attack:

CIFAR-10

AllConv



SHIP
CAR (99.7%)

NiN



HORSE
FROG (99.9%)

VGG16



DEER
AIRPLANE (85.3%)

ImageNet

BVLC AlexNet



Cup (16.48%)
Soup Bowl (16.74%)



Bassinet (16.59%)
Paper Towel (16.21%)

Performance of the one pixel attack

One pixel attack it is often performed using search algorithms (like Differential Evolution (DE)) to find the most effective

- pixel position to change and
- the changed color value.

Approaches of Global Explanations

Collection of Local explanation

Global Explanations from Local Feature Importances:

Local method explains a single prediction

local behavior for a single instance

Can't examine all explanations

Instead pick k explanations to show to the user

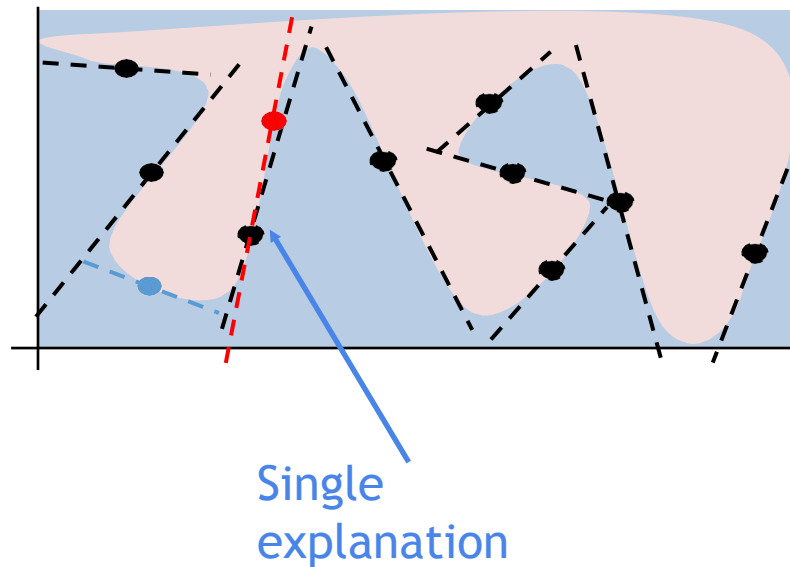
Representative

Should summarize the model's global behavior

Diverse

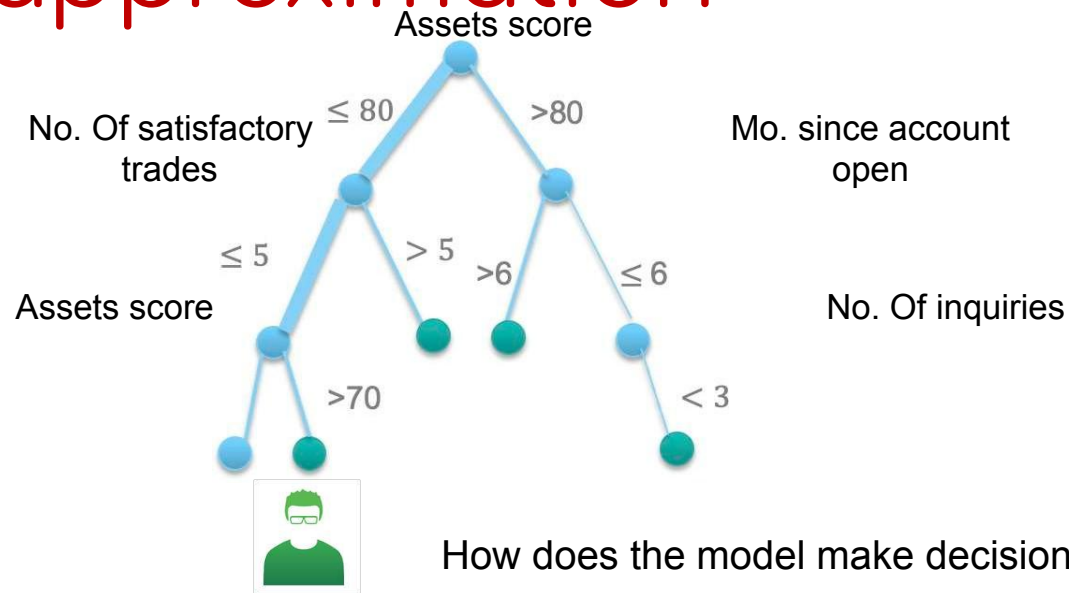
Should not be redundant in their descriptions

submodular optimization and *greedily* picks k explanations



Model distillation

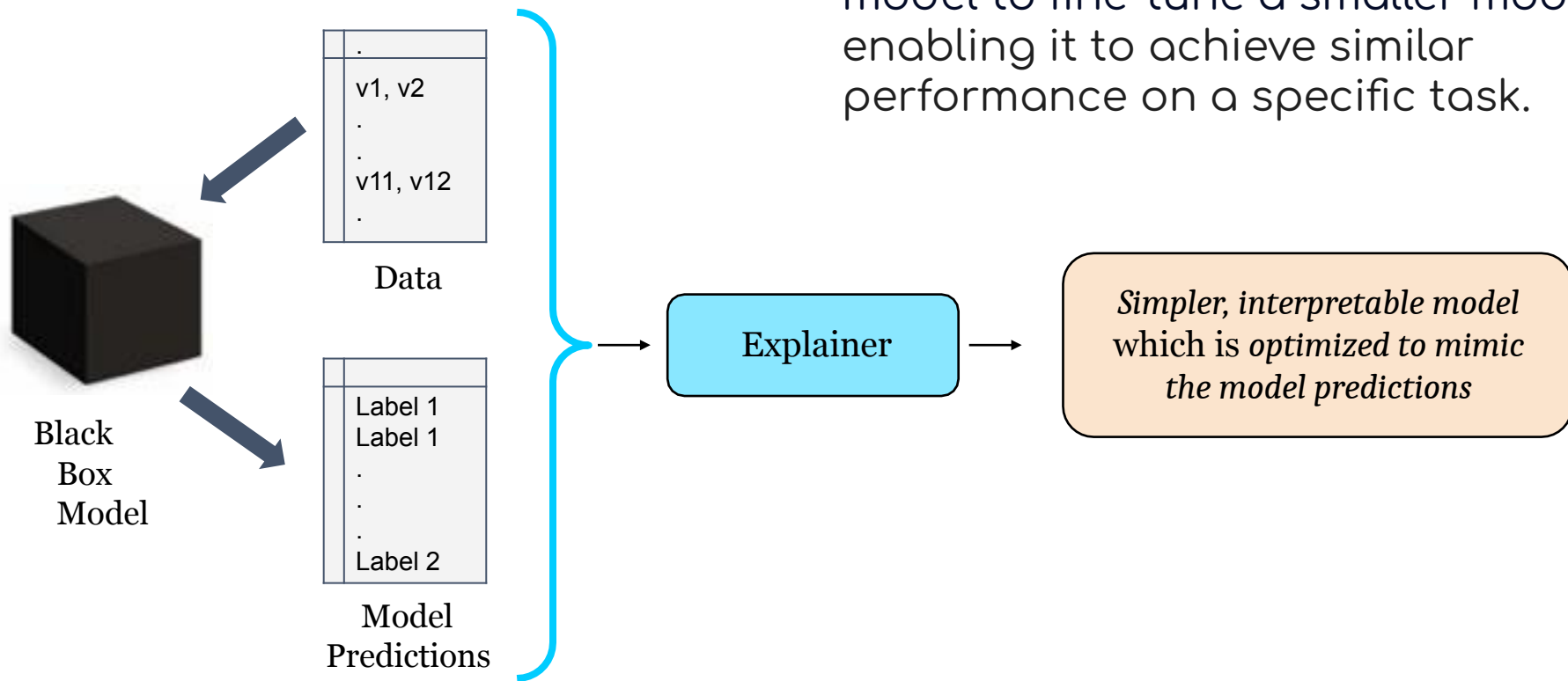
Explaining the model: decision-tree approximation



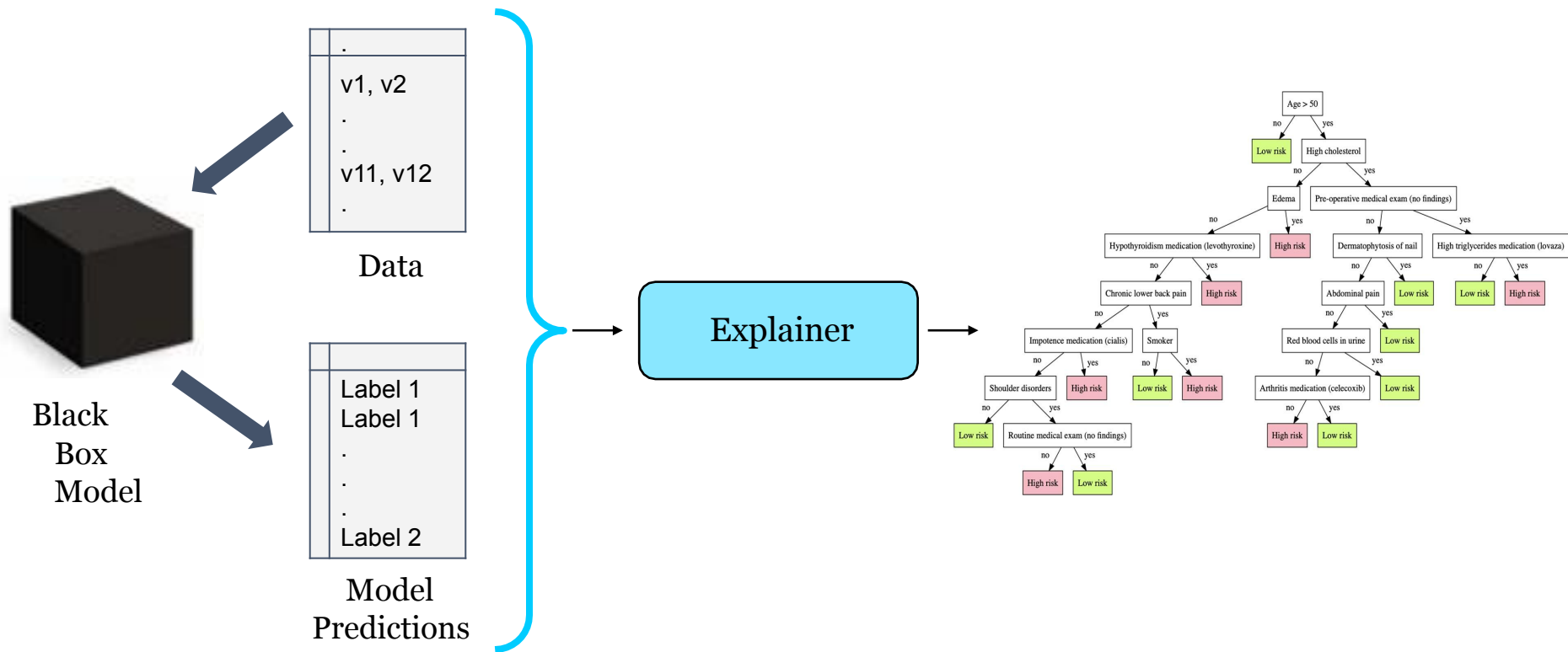
Data Scientist

Model Distillation

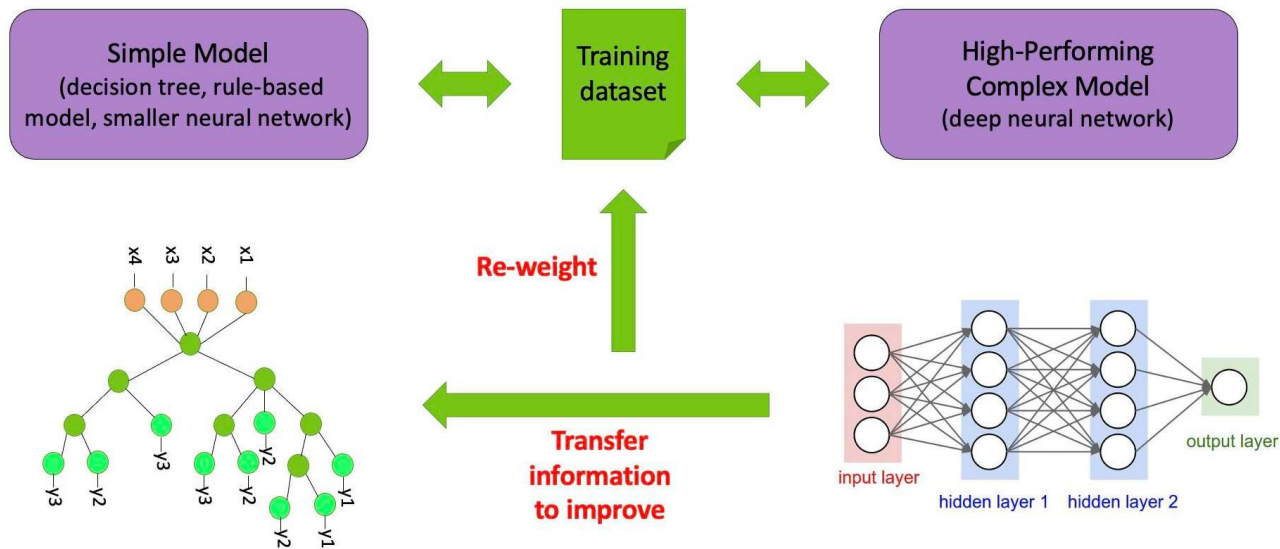
Model Distillation allows you to leverage the outputs of a large model to fine-tune a smaller model, enabling it to achieve similar performance on a specific task.



Model Distillation Using Decision Trees



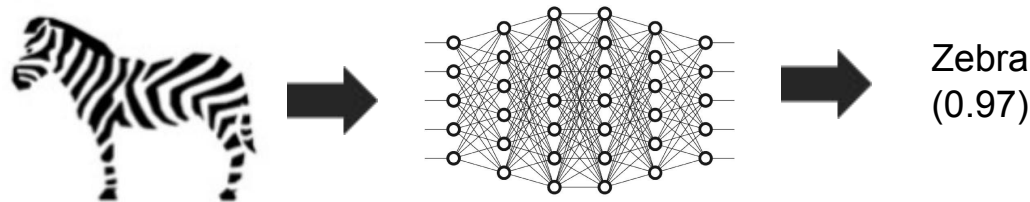
Knowledge distillation (approximation)



The goal of knowledge distillation is to train a more compact model to mimic a larger, more complex model.

Representation based

Representation Based Explanations

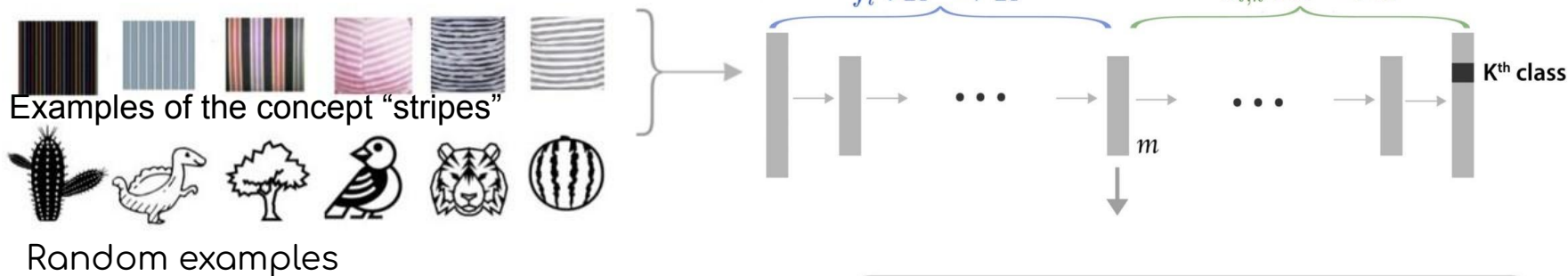


Example:

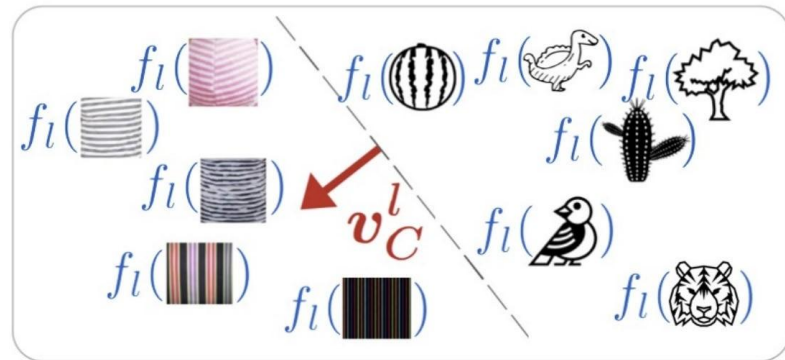
How important is the notion of “stripes” for this prediction?

Representation Based Explanations

Concept Activation Vectors (TCAV)



- Train a linear classifier to separate activations
- The vector orthogonal to the decision boundary denotes the concept "stripes"
- Compute gradient w.r.t. this vector to determine how important is the notion of stripes for a prediction

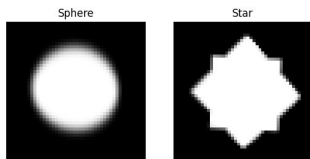


Example: classification of nodule in lungs



Calibration concepts sphere and star

- Cluster A - star
- Cluster B - spheres

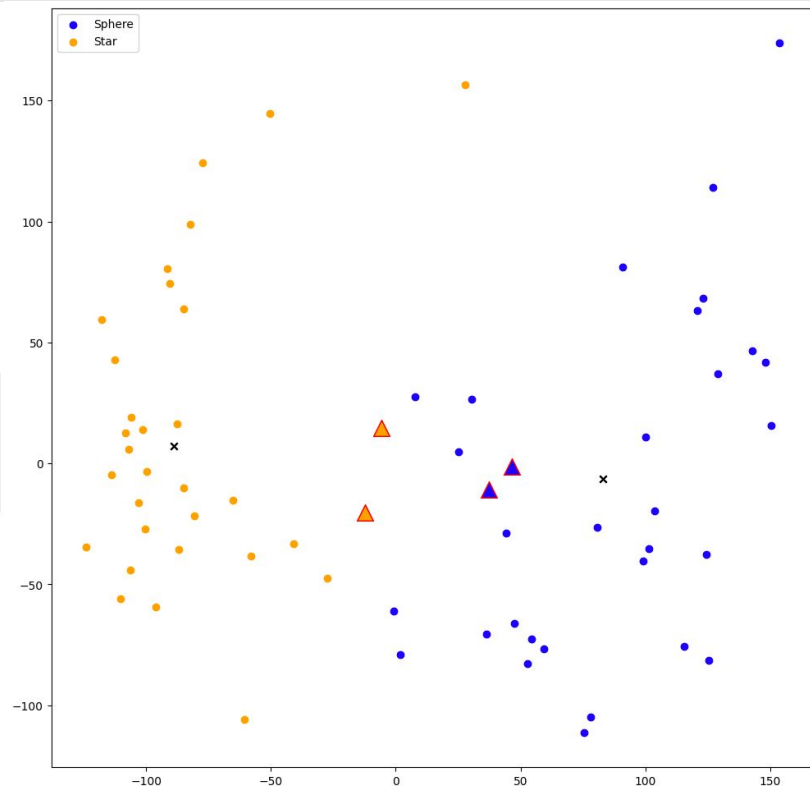
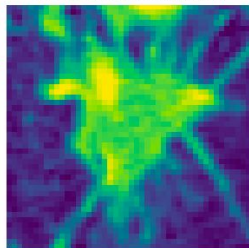
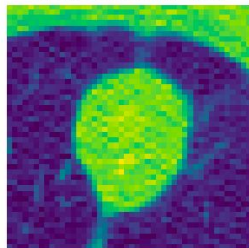
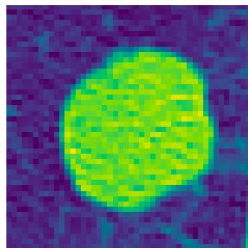
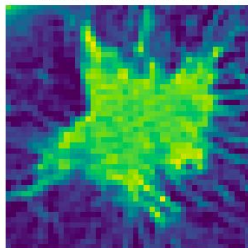


centroid A distance: 83.41
centroid B distance: 91.44

centroid A distance: 127.42
centroid B distance: 46.12

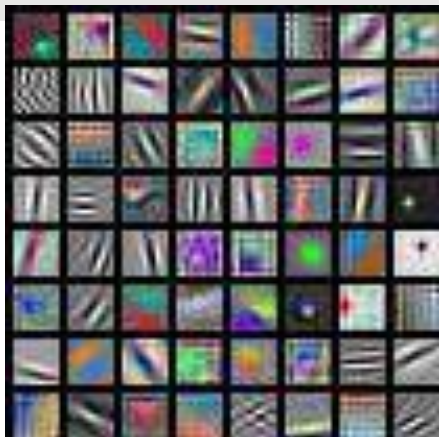
centroid A distance: 135.83
centroid B distance: 36.85

centroid A distance: 81.17
centroid B distance: 96.56



Visualizing Neural Networks (Global approach)

First Layer: Visualize Filters



AlexNet:
 $64 \times 3 \times 11 \times 11$



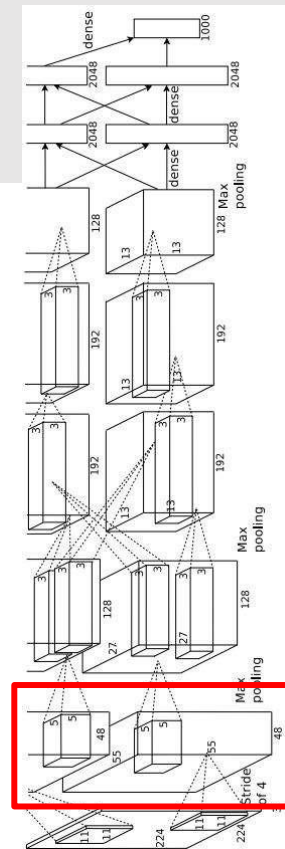
ResNet-18:
 $64 \times 3 \times 7 \times 7$



ResNet-101:
 $64 \times 3 \times 7 \times 7$



DenseNet-121:
 $64 \times 3 \times 7 \times 7$

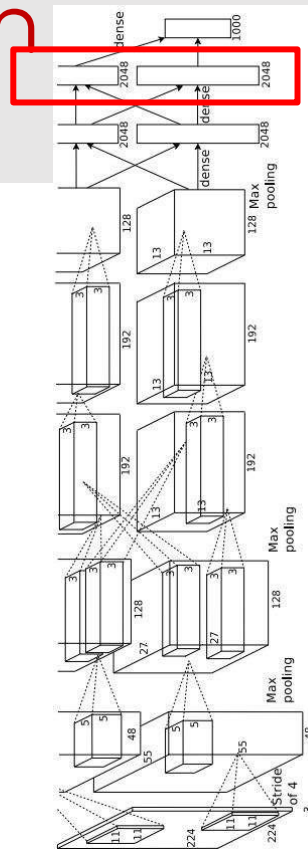
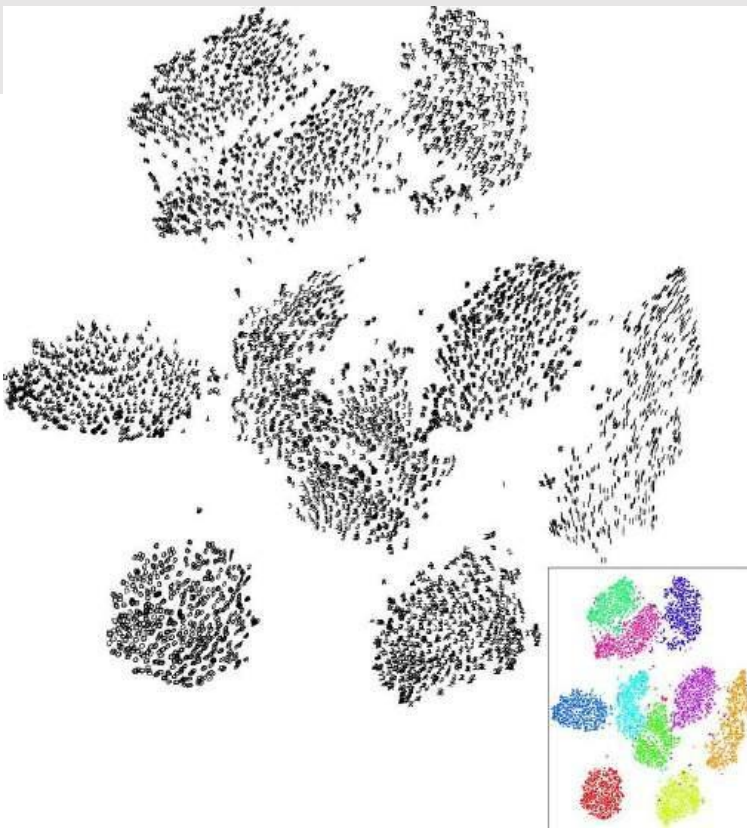


Krizhevsky, "One weird trick for parallelizing convolutional neural networks", arXiv 2014
He et al, "Deep Residual Learning for Image Recognition", CVPR 2016
Huang et al, "Densely Connected Convolutional Networks", CVPR 2017

Last Layer: Dimensionality Reduction

Visualize the “space” of Fully Connected Layer feature vectors by reducing dimensionality of vectors from 4096 to 2 (or more) dimensions

Simple algorithm:
Principle Component Analysis (PCA)

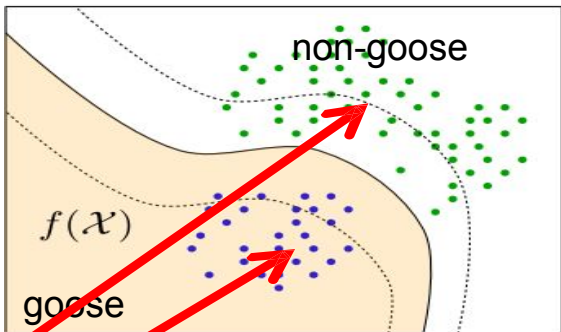


Van der Maaten and Hinton, “Visualizing Data using t-SNE”, JMLR 2008
Figure copyright Laurens van der Maaten and Geoff Hinton, 2008. Reproduced with permission.

Class Prototypes

“How does a goose typically look like according to the neural network?”

...maximize the activation functions of a certain layer or neuron by changing the input picture.

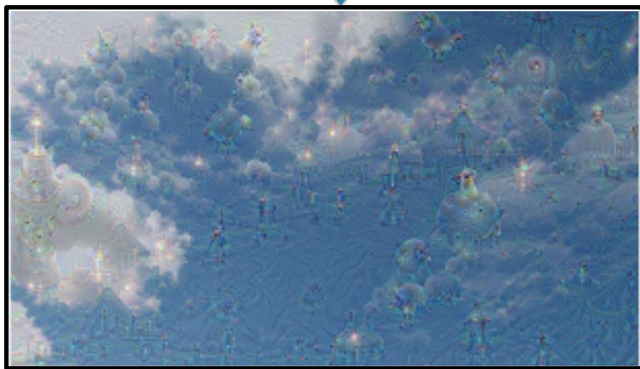
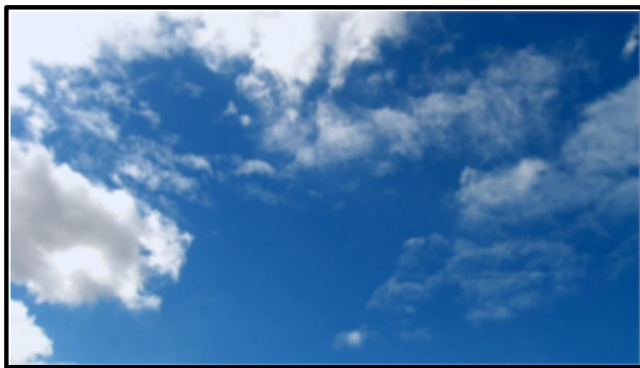


Class prototypes

$$\arg \max_x f(x) + \text{reg.}$$



Deep Dream: Going Deeper into Neural Networks



Deep Dream is a feature visualization technique that synthesizes images that **strongly activate network layers**.

By visualizing these images, you can highlight the image features learned by a network. These images are useful for understanding and diagnosing network behavior.

The dreaming idea can be applied to **hidden (internal)** neurons



"Admiral Dog!"



"The Pig-Snail"



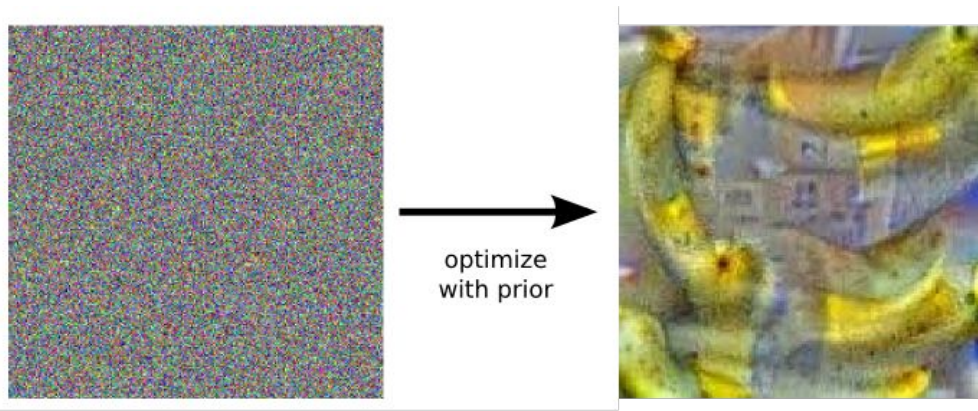
"The Camel-Bird"



"The Dog-Fish"

Going Deeper into Neural Networks

Say you want to know what sort of image would result in “Banana.” Start with an image full of random noise, then gradually tweak the image towards what the neural net considers a banana.



Example



dumbbells

Has the NN learned the correct flags for object detection and classification?

Although the network can detect them, it did not derived the basic essential!

Metrics and tools for XAI evaluation

Metrics for XAI evaluation

Metrics of inherent explainability

- Fidelity/faithfulness
- Stability
- Compactness
- ...

User-dependent measures

- Comprehensibility
- Explanation satisfaction
- ...

Task oriented measures

- Task performance
- Impact on AI interaction
 - Trust (calibration) in model
- Task or AI system satisfaction

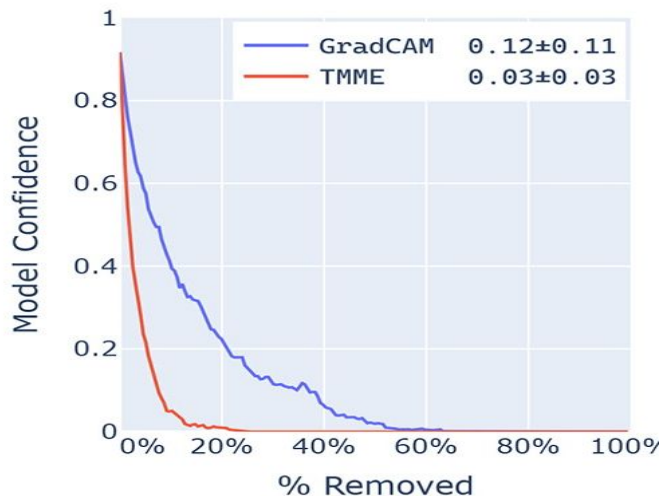
XAI evaluation

Faithfulness

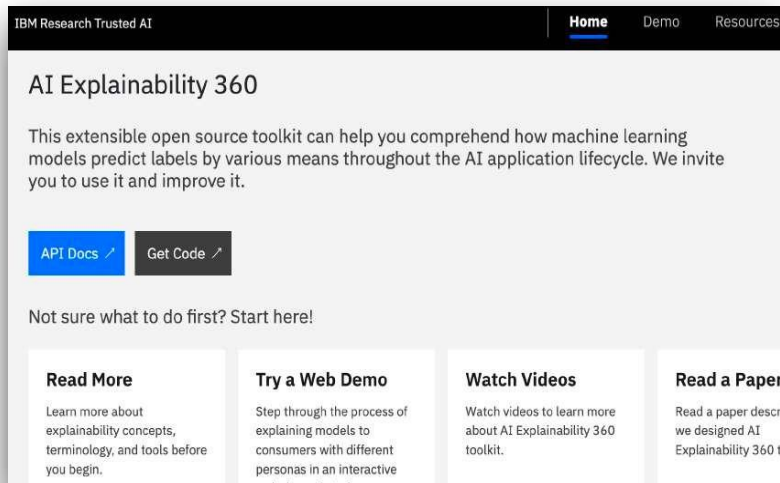
- Computes which feature has the most impact on the model output when individually changed
- Removing each important feature should result in decreasing the model performance

Saliency Map Evaluation

Saliency map performance comparison example: transformer multimodal explainability (TMME) performed better than gradient-weighted class activation mapping (GradCAM).

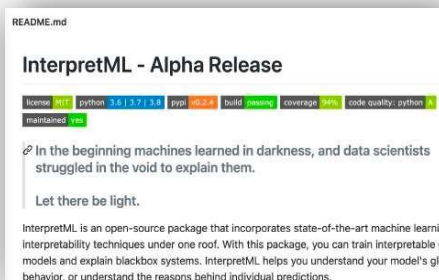


XAI open-source toolkits

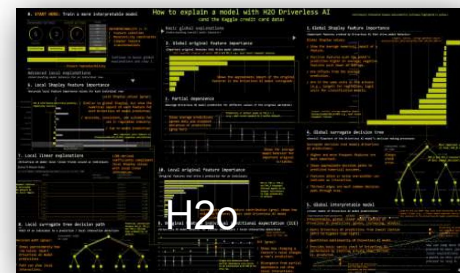


AIX 360

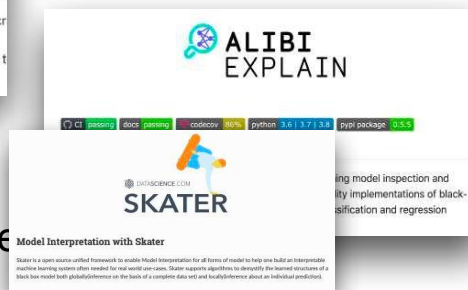
<http://aix360.mybluemix.net/>



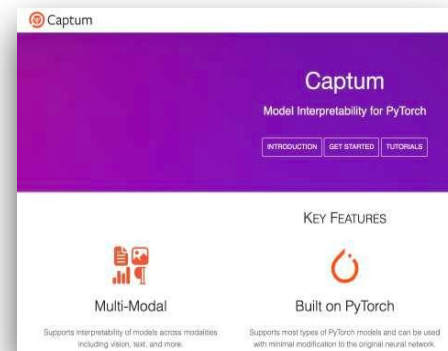
Microsoft Interpret



She



Oracle Skater



PyTorch Captum

Lucid: a neural network visualization library

Lucid is a collection of infrastructure and tools for research in neural network interpretability

For instance, by combining feature visualization and class attribution we can start to answer complex questions like, "How does a neural network decide between a **Labrador retriever** and a **tiger cat**?"



Channels that most support

Labrador retriever

Channels that most support

tiger cat

feature visualization

hover for attribution maps →

net evidence	0.53	0.49	0.39	0.43	0.50	0.56
for "Labrador retriever"	0.40	0.40	0.43	-0.23	-0.40	-0.14
for "tiger cat"	-0.13	-0.09	0.04	0.20	0.10	0.42

End-User-Centered Explainable AI

Human-Centered Explainable AI (XAI)

Explainability from a user and usage perspective

XAI literature generally divides its users into two groups according to their level of technical knowledge in AI:

- **Technical users - AI developers**
 - the primary focus of current XAI research,
 - debugging, understanding, and improving AI models for technical users, such as data scientists, AI researchers, and developers...
- **Non-technical users - stakeholders**
 - Domain experts / end users

Examples: drivers of autonomous driving vehicles, or domain experts-such as doctors using AI-assisted technology in diagnostic tasks, judges using AI to support reaching a guilty verdict, and bankers using AI to assist in approving loan applications.

Question-Drive Design Process for XAI Question Bank

LIAO, Q. Vera; GRUEN, Daniel; MILLER, Sarah. Questioning the AI: informing design practices for explainable AI user experiences. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. 2020. S. 1-15.



Question-Drive Design Process for XAI UX

Mapping guide between question categories and candidate XAI techniques, with exemplary descriptions of explanations provided by these techniques and links to code libraries

Question	Explanations	Example XAI techniques
Global how (global model-wide)	- Describe the general model logic as feature impact⁺, rules⁺ or decision-trees⁺ (sometimes need to explain with a surrogate simple model) - If a user is only interested in a high-level view, describe what are the top features or rules considered	ProfWeight ⁺⁺ , Global Feature Importance ⁺ , PDP ⁺ , BRG ⁺ , GLRM ⁺ , Rule List ⁺ , DT Surrogate ⁺
Why	- Describe what key features of the instance determine the model's prediction of it⁺ - Describe rules⁺ that the instance fits to guarantee the prediction - Show similar examples⁺ with the same predicted outcome to justify the model's prediction	LIME ⁺ , SHAP ⁺ , LOCO ⁺ , Anchors ⁺ , ProtoDash ⁺
Why not	- Describe what changes are required for the instance to get the alternative prediction and/or what features of the instance guarantee the current prediction⁺ - Show prototypical examples⁺ that had the alternative outcome	CEM ⁺ , Prototype counterfactual ⁺ , ProtoDash ⁺ (on alternative class)
How to be that (a different prediction)	- Highlight features that if changed (increased, decreased, absent, or present) could alter the prediction⁺ - Show examples with minimum differences but had a different outcome than the prediction⁺	CEM ⁺ , Counterfactuals ⁺ , DiCE ⁺
How to still be this (the current prediction)	- Describe features/feature ranges⁺ or rules⁺ that could guarantee the same prediction - Show examples that are different from the particular instance but still had the same outcome	CEM ⁺ , Anchors ⁺
What if	Show how the prediction changes corresponding to the inquired change	PDP , ALE , What-if Tool
Performance	- Provide performance metrics of the model - Show uncertainty information for each prediction - Describe potential strengths and limitations of the model	Precision, Recall, Accuracy, F1, AUC Uncertainty Quantification 360 FactSheets , Model Cards
Data	Document comprehensive information about the training data, including the source, provenance, type, size, coverage of population, potential biases, etc.	FactSheets , DataSheets
Output	- Describe the scope of output or system functions. - Suggest how the output should be used for downstream tasks or user workflow	FactSheets , Model Cards

Liao QV, Pribić M, Han J, Miller S, Sow D.
Question-driven design process for explainable AI
user experiences. arXiv preprint arXiv:2104.03483. 2021
Apr 8.

Recommended links & literature

Recommended links & literature

Machine Learning Explainability Workshop | Stanford

<https://www.youtube.com/playlist?list=PLoROMvodv4rPh6wa6PGcHH6vMG9sEIPxL>
[Stanford Seminar - ML Explainability Part 3 | Post hoc Explanation Methods](#)

[Vera Liao | Introduction to Explainable AI: Techniques and Design](#)

LIAO, Q. Vera; GRUEN, Daniel; MILLER, Sarah. Questioning the AI: informing design practices for explainable AI user experiences. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. 2020. S. 1-15.

Heat-Mapping website

<http://www.heatmapping.org/>

More resources for XAI

Toolkits/Libraries/Readings/Design guidelines

- [Microsoft Interpret](#)
- [PyTorch Captum](#)
- [Interpretable ML e-book](#)
- [Google PAIR: Explainability+Trust](#)
- [SAP Deelines for Explainability](#)
- [IBM Design for AI: Explainability](#)
- [UXAI for Designers](#)
- [Lingua Franca: Transparency](#)